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Find right countenance for your input—Improving automatic emoticon recommendation system with distributed representations

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ABSTRACT

Emoticons are popularly used to express user's feelings in social media, blogs, and instant messaging. However, the number of emoticons existing in emoticon dictionaries which users select from is large, thus, it is difficult for users to find the desired emoticon that matches the content of their messages. In this paper, we propose a method that supports users' emoticon selection by reordering 167 unique emoticons in the emoticon dictionary by applying pre-trained models learned from large data in Japanese. We evaluated whether adapting a pre-trained model to our emoticon recommendation system achieves better results than just learning surface patterns of text and emoticon. We collected sets of sentences and emoticons in Japanese from the Internet and pre-trained models (i.e. Word2vec, ELMo, and BERT) that learned from large Japanese textual data and used deep learning techniques such as BiLSTM and fine-tuning for learning. We confirmed that fine-tuning our data with BERT achieved the best recommendation accuracy of 52.98%, recommending the correct emoticon within the top 25 (top 15%) of the emoticons. Moreover, we confirmed our intuition that widely used Wikipedia-based pre-trained models are not the best voice for the facemark recommendations.

1. Introduction

Emoticons are widely used in computer-mediated communication (CMC), with instant messaging and social media as the most prominent examples. Unlike emojis which are pictograms that can represent objects other than faces (Fig. 1, on the left), emoticons represent facial expressions and are composed of characters and symbols (Fig. 1, on the right). There are two types of emoticons: vertical style (e.g. ":)"), mainly used in Western countries, and horizontal style (e.g."(‾‿‾)"), mainly used in the Far East (Park, Barash, Fink, & Cha, 2013). While the vertical style emoticons are composed of symbols and Latin letters, and are rotated by 90 degrees, the horizontal style is also composed of symbols from other writing systems like Cyrillic alphabet, Chinese ideograms, or Japanese and Korean syllabaries (Park et al., 2013; Tomi, Marinez, & Vrbanec, 2013). Moreover, the horizontal style is not rotated 90 degrees which makes it more easily comprehensible to a reader. Since the horizontal style consists of a larger number of symbols and letters than the vertical one, over one million examples of the latter were discovered in the Internet[1], while only around 300 vertical-style emoticons

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[1] Available in Flick (https://kaomoji.tokyo/), an input method application with a dictionary comprising over 2 million emoticons.

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Fig. 1. Difference between emojis and emoticons.

are listed by Wikipedia.[2]

CMC may cause more misunderstandings than face-to-face communication because it cannot convey attitudes via nonverbal cues such as facial expression, body movement, or tone of voice (Jibril & Abdullah, 2013). Nonverbal cues help humans to understand feelings and intentions of interlocutors and may take up an estimated 93% of communication (38% feelings and attitudes from voice, 55% from facial expression) (Mehrabian, 1971). The role of using emoticons is to express users' feelings and intentions, and to convey their messages more clearly. When emoticons are inserted after a neutral input, they serve as an emotive indicator (Yamamoto, Kumamoto, & Nadamoto, 2015). Emoticons can enhance the emotive meaning of a message when a text is followed by an emoticon of the same polarity (Derks, Bos, & von Grumbkow, 2007; Inoue, Fujimaki, & Ishizaki, 1997; Ip, 2002). Moreover, emoticons can weaken the negative emotion of the message when a positive emoticon is added to a negative message (Harada, 2004; Yamamoto et al., 2015).

Users nowadays insert emoticons into their typed texts by selecting a desired one from an internal dictionary on their smartphones or tablets (Fig. 2, left). On computers, it is more common to insert emoticons by using input method editor (IME) and convert the word "かおもじ" (*kaomoji*,[3] meaning "facemarks" in Japanese) to one that users want to input into their messages (Fig. 2, right). The number of emoticons registered in Japanese keyboard on smartphones and tablets is 343 in iOS 13.3.1 and 386 in Android 10 and it has been continually increasing from the previous OS versions which included around 200 emoticons (Urabe, Rzepka, & Araki, 2014). Since these numbers are large and the emoticons' expressions are diverse, it is difficult for users to find an emoticon that they consider appropriate for their messages. Thus, a method to assist users with choosing emoticons that match their inputs is desirable. The current function that assists users with selecting an emoticon displays a set of previously chosen ones on the top of the list and does not take the content of a message into account when recommending (Urabe et al., 2014).

The aim of our research is to support users' emoticon selection by recommending facemarks while considering the semantics of the input text message. In this paper, we propose an automatic emoticon recommendation method based on the context of the message by applying models pre-trained on large data in Japanese. The main contribution of this paper is as follows:

- We quantitatively confirmed that adapting latent semantics learned from a large corpus to our emoticon prediction model increases the recommendation accuracy to a greater degree than only learning surface patterns of text and emoticons as previous methods did. Concretely, a method applying pre-trained models achieved higher recommendation accuracy than the conventional methods by 0.60 pt to 23.45 pt (details are written in Section 6). We adapted Word2vec, ELMo, and BERT to our emoticon prediction model and empirically compared them with conventional methods (e.g. Bag-of-words or n-gram of words and characters with SVM classifier, see Section 5.2).

The outline of this paper is as follows: in the next section, we present related research and describe some differences between several studies performed so far. In Section 3, we introduce our preliminary study that aimed at understanding the usage of emoticons in messages. We show the results of our investigation listing problems of recommending emoticons with the rule-based methods (details of the rule-based methods are written in Section 2). Section 4 describes our proposed methods, Section 5 details an evaluation experiment, and Section 6 shows which pre-trained model works best and how differently these models perform. Finally, conclusions and future work are presented in the last section.

[2] List of emoticons: https://en.wikipedia.org/wiki/List_of_emoticons
[3] Italic font indicates foreign language transcripts, technical terms, key terms, or labels.



Fig. 2. Way to insert emoticons from a smartphone/tablet (left) and a computer (right).

2. Related research

Rule-based emoticon recommendation methods were proposed by Suzuki and Tsuda (2006), Emura and Seki (2012), and Urabe et al. (2014). Suzuki and Tsuda (2006) proposed a method that analyzes an emotive load in an input sentence and recommends emoticons based on the emotion type of the emotive expression. In their research, they used two lexicons: one containing emotion-expressing phrases categorized into emotion types, and one comprised of emoticons accompanied by emotive phrases. The emoticon lexicon was created manually which aided to choose a representative emoticon to each emotive phrase. Emura and Seki (2012), proposed a method that recommends emoticons not only from the emotive phrases found in an input sentence but also from communication-related (e.g. greetings, congratulating, etc.) and activity-related expressions (e.g. sleeping). In their research, they constructed a corpus of 3975 sentences that contains emotional phrases, communication-related and activity-related expressions. The expression types were annotated manually. Emoticon dictionary was also created by analyzing the corpus and registering a set of emoticons, an expression type, and emoticon occurrence in the corpus. Their method recommended emoticons by predicting the expression type from the input by extracting similar texts from the corpus using k-nearest neighbors algorithm and then choosing frequently used emoticons that appeared in corpus sentences of the same expression type. Urabe et al. (2014) proposed a method that recommends emoticons based on the emotion type of an emotive expression extracted from an input, emotive strength of an emoticon, and user's selection frequency of an emoticon. They applied ML-Ask (Ptaszynski, Dybala, Rzepka, Araki, & Masui, 2017) to analyze the emotion type of an input and created an emoticon dictionary by manually annotating a degree of emotive load for each of 10 emotion types (*Joy, Like, Relief, Anger, Dislike, Fear, Sadness, Amazement, Excitement, Shyness*) (Nakamura, 1993), and recording which emoticon was selected when an input contained a particular emotion type. It should be underlined that all the above methods (Emura & Seki, 2012; Suzuki & Tsuda, 2006; Urabe et al., 2014) can only recommend emoticons when there are emotive expressions (and communication and activity-related expressions for method presented in Emura & Seki (2012)) discovered in the input sentence which means that these methods cannot recommend emoticons if the sentence does not contain emotion-related phrases. In reality, however, there are input messages with emoticons that do not contain emotive expressions (see details in Section 3). Moreover, previous methods (Emura & Seki, 2012; Suzuki & Tsuda, 2006; Urabe et al., 2014) recommend emoticons that express the same polarity/emotion of the emotive expression contained in the input although polarity and emotion type of emotive expression in a sentence and emoticon do not always match with each other (more details in Section 3). In our study, we aim at building a novel system that can recommend suitable emoticons by understanding the emotion from context, even if there are no emotion-related words present. Direct comparison with the aforementioned rule-based methods is not possible because our method covers a much broader range of inputs not containing emotive phrases. However, we introduce frequency of emotive phrase-based baseline which we believe is similar to the algorithm proposed by Emura and Seki (2012).

Emoji prediction competition (Barbieri et al., 2018) took place in 2018 where participants were asked to predict emojis to be most likely used along with the text in English and Spanish tweets. The dataset used in this challenge comprised of 550 K tweets (500 K for training data, 50K for test data), each containing one of the 20 most frequent emojis. The top F-scores of 35.99% and 22.36% were achieved by Çöltekin et al.'s method (Çöltekin & Rama, 2018) for English and Spanish, respectively. Çöltekin et al.'s method used bag-of-N-grams features of both words and characters with Support Vector Machine (SVM) classifier. Recently, Zhang, Zhou, Erekhinskaya, and Moldovan (2020) outperformed Çöltekin et al.'s F-score result by 2.53 points by fine-tuning with BERT (Devlin, Chang, Lee, & Toutanova, 2019). In their research, they stated that applying knowledge learned from other natural language processing tasks improves the results because such a model can utilize semantic information (for instance the relation between words and sentences) in the new task. As mentioned before, the difference between emojis and facemarks is that the former expresses not only emotion with facial expressions but also objects, states, and so on. The latter, on the other hand, mostly expresses emotions with various kinds of

facial appearance. As the emoji dataset introduced in Barbieri et al. (2018) included both faces and objects (Fig. 3) and the F-score of facial emoji prediction results tends to be lower than that of objects by the average 23.65 pt for English and 1.57 pt for Spanish, it becomes quite obvious that recommending emoticons for text input by applying machine learning techniques is a challenging task. Moreover, to the best of our knowledge, no studies are utilizing a machine learning approach to recommend desired emoticons for an input text. In this paper, we propose a method for recommending emoticons to users by reordering a set of 167 unique emoticons based on the content of the text even if it does not contain any emotive words.

3. Preliminary investigation

First, we investigated how emoticons are used in texts and what are the limitations of the rule-based methods (Emura & Seki, 2012; Suzuki & Tsuda, 2006; Urabe et al., 2014) by analyzing data containing pairs of Japanese sentences and emoticons. As the rule-based methods recommend emoticons of the same polarity/emotion based on the emotive words written in inputs, we performed two analyses: (1) the percentage of inputs that contained emotive phrases and (2) the percentage of data where the polarity and emotion type of emotive phrases matched with those of the inserted emoticons.

3.1. Data description

The data contained 652,924 pairs of sentences and emoticons and was collected from Twitter,[4] Nihongo Web Corpus (NWC) (Yata, 2010), and YACIS (Ptaszynski, Dybala, Rzepka, Araki, & Momouchi, 2012). Examples are in Table 1. We used these data because emoticons occur more frequently in casual messages on social media and in blogs. The Twitter data (Japanese tweets) was collected between March 16, 2010 and January 10, 2011. NWC is text data collected via Yahoo! Web search API[5] between June and September in 2010. YACIS data was collected from Ameba blog[6] in December, 2009. Some data contained multiple sentences merged into one line, therefore, we divided the text into sentences by periods, exclamation marks, question marks, emoticons, and emojis. We only retrieved data that contained at least five tokens (POS chunks) of natural language text followed by one emoticon. We did not collect sentences that contained less than five tokens because we considered them as too short and therefore difficult to have any emoticons suggested. The average number of tokens in one text was 10.3 and the longest sentence contained 261 tokens.

We also automatically annotated polarity and emotion types in the data when there were emotive phrases in the sentence by using ML-Ask (Ptaszynski et al., 2017). The system recognizes polarity and emotion category in text by matching emotive phrases appearing in the input based on emotive expression dictionary (Nakamura, 1993). Ten emotion types used in ML-Ask (*Joy, Like, Relief, Anger, Dislike, Fear, Sadness, Amazement, Excitement, Shyness*) are also borrowed from (Nakamura, 1993). The polarity in ML-Ask is divided into Positive, Negative, and Neutral categories. Emotion types are grouped as follows: *Joy, Like, Relief* belong to Positive, *Anger, Dislike, Fear*, and *Sadness* to Negative, and *Amazement, Excitement*, and *Shyness* are categorized as Neutral categories. However, words expressing *Amazement, Excitement*, and *Shyness* can be either positive or negative depending on the context, therefore, we annotated them as Other instead of Neutral. When there were no emotive phrases in a sentence, polarity and emotion categories were marked as None. If there are multiple emotive phrases in a given input, we only used polarity and emotion type of the last matched emotive expression because in Japanese language rightmost ones tend to influence the selection of the emoticon most (Emura & Seki, 2012).

We used emoticons registered in an emoticon dictionary created by Urabe, Rzepka, and Araki (2016). It contains 299 different emoticons manually annotated with a strength between 0 and 1 for each ten emotion type defined in Nakamura's emotive expression dictionary (Nakamura, 1993). Examples are given in Table 2. The polarity and the emotion type of each emoticon are determined by using the highest value among the ten emotion types. For example, "(*^‿^*)" was annotated as Positivefor polarity type and *Joy* for emotion type because *Joy* had the highest emotional load (see Table 2, second row). We used annotated emoticons to search the aforementioned corpora and we were able to collect sentences with 167 out of 299 emoticons. The matched 167 emoticons are listed in Table 3.

3.2. Results

We confirmed that 490,345 (75.1%) sentences do not contain emotive phrases and 162,579 (24.9%) sentences contain emotive phrases (Fig. 4). This indicates that the rule-based methods (Emura & Seki, 2012; Suzuki & Tsuda, 2006; Urabe et al., 2014) cannot recommend emoticons for 75.1% of the messages. For 162,579 (24.9%) sentences containing emotive phrases, there were 125,362 (19.2%) messages with emoticons expressing the same polarity as the emotive phrases and 37,217 (5.7%) messages without emoticons of the same polarity as emotive phrases. This means that at least 37,217 (5.7%) inputs cannot be accurately recommended by the rule-based methods (Emura & Seki, 2012; Suzuki & Tsuda, 2006; Urabe et al., 2014).

Fig. 5 shows the percentage of emotion type of inserted emoticons for each emotion type of sentences. From Fig. 5, we can see that nearly 20% of sentences containing positive (*Joy, Like, Relief*) expressions did not co-occur with emoticons of the same polarity. For messages containing negative (*Anger, Dislike, Fear, Sadness*) expressions, 59.3% (*Anger*), 51.1% (*Dislike*), 53.6% (*Fear*), and 29.7% (*Sadness*) of inserted emoticons did not express negative emotion but expressed either positive or other emotion types. For inputs

[4] Twitter: https://twitter.com/
[5] Yahoo! Web search API: https://developer.yahoo.co.jp/changelog/2013-08-14-search.html. The service ended on August 14, 2013.
[6] Ameba blog: https://ameblo.jp/



Fig. 3. Emojis used in SemEval 2018 Task 2, listed in frequency order, borrowed from (Barbieri et al., 2018).

Table 1
Examples of sentence-emoticon pairs. Emotive words are emphasized in bold.

Text	Emoticon	Text polarity	Text emotion	Emoticon polarity	Emoticon emotion
僕もイベントが好きです(*I also* ***like****events*)	(‾‾)	Positive	*Like*	Positive	*Joy*
もう一回順番待ちするの面倒くさいなあ (*It is* ***troublesome****to wait for my turn again)*	(;_;)	Negative	*Dislike*	Negative	*Sadness*
野宿できるとこ探さないと (*I have to look for a place to camp out*)	(ーー；)	None	None	Negative	*Dislike*

Table 2
Example data from emoticon dictionary (Urabe et al., 2016). The strongest emotion type for each emoticon is emphasized in bold.

Emoticon	*Joy*	*Like*	*Relief*	*Anger*	*Dislike*	*Fear*	*Sadness*	*Amazement*	*Excitement*	*Shyness*
(*^_^*)	**0.87**	0.78	0.65	0.10	0.08	0.06	0.09	0.10	0.16	0.25
(T_T)	0.08	0.08	0.12	0.20	0.51	0.47	**0.87**	0.19	0.25	0.29
(;°0 °)	0.08	0.07	0.07	0.15	0.33	0.39	0.18	**0.95**	0.48	0.32

containing *Amazement, Excitement*, and *Shyness* expressions, 72.3% (*Amazement*), 81.3% (*Excitement*), and 71.7% (*Shyness*) of inserted emoticons expressed either positive or negative emotion. These results suggest that the rule-based methods (Emura & Seki, 2012; Suzuki & Tsuda, 2006; Urabe et al., 2014) cannot recommend desired emoticons to the aforementioned percentage of sentences due to the mismatch of the emotion types between sentences and emoticons.

Examples of sentence-emoticon pairs with unmatched polarity are shown in Table 4. The first example in Table 4 contains a word ``笑う" (*warau*, meaning "laugh") which has a positive meaning. However, in this context, the sender is being laughed at. Thus, we assumed that emoticon "(/ _ ;)", which expresses *Sadness*, was inserted to express the sender's negative feeling of being ridiculed. The second example in Table 4 contains an emotive word ``恥ずかしさ" (*hazukashisa*, "shyness" or "embarrassment"). It is then followed by a sentence ``消去しました" (I deleted it), therefore, we assumed that the sender had a negative feeling toward the message and added the negative emoticon ``( ; ; )". The third example contains an emotive word ``泣いて" (*naite*, "cry"), however, this word is rather not about the sender but towards someone the sender tries to comfort. Thus, we assumed that the sender wants to express a positive feeling to cheer up the receiver.

From the results presented in Section 3.2, we concluded that it is not sufficient to recommend emoticons solely from the emotive word in a sentence because many inputs do not contain emotive phrases. Moreover, the polarity/emotion type between emotive expressions and emoticons do not always match due to stronger relationship to neighboring non-emotive words. Therefore, we propose a method that considers all words of the input message and can understand their context by adapting pre-trained models to machine learning techniques.

4. Proposed system

The procedure of the proposed system is as follows. The numbers below correspond to the numbers in Fig. 6.

1. A user (U) inputs a sentence.
2. The system (S) reorders emoticons in the emoticon dictionary according to the result of the prediction model and displays the emoticon list to the user.
3. The user (U) selects an emoticon from the list.
4. The system inserts the selected emoticon after the user's input.

Table 3
167 emoticons and their emotion and polarity types.

Polarity	Emotion	Quantity	Emoticon				
Positive	*Joy*	60	(≧ ω≦)	(ﾉ∇≦ *)	ヾ (o´∀ ` o)ﾉ	(⌒▽⌒)	☆ (≧∀≦ *)ﾉ
			Σ d(≧∀≦ *)	(￣▽￣)	(*≧ m≦ *)	(^。^)	(=^ェ^=)
			。・゜・ (ﾉ∀ ‘)・゜・。	(^o^)	(*^▽^*)	(#^.^#)	(o^ω^o)
			(*^_^*)	(* ´艸｀)	(○ ´艸｀)	(^з^)-☆	(*ﾉ∀ ‘*)
			(((o(*°▽°*)o)))	(^ー°)	(*^○^*)	(⌒⌒)	(≧艸≦ *)
			(≧ з≦)	(^O^)	(^ ω ^)	(=^x^=)	＼(^o^)／
			(*⌒⌒*)	(ﾋﾒ｀∀ ´) ﾋﾒ	ヾ (● ´∀ ` ●)	(^○^)	(^∇^)
			(^o^)	(^_^)	(● ` ・ω・)	d(°∀°*)	(´艸｀)
			.+(´ ^ ω ^ ‘)+.	(´艸‘*)	(^ O ^)	(● ´∀ ` ●)ﾉ	o(≧ω≦)o
			(` ∀ ´)	(*^o ^*)	(^∀ ^)	(^ω^)	(ﾉ°∀°)ﾉ
			(°∇^*)	(⌒⌒)	(*´∀ ` *)	(^-^)	ヽ (≧∀≦)ﾉ
			(^人^)	☆*:.。. o(≧▽≦)o .。.:*☆		♪———O (≧∇≦) O———♪	
			(^◇^)	*・゜ ゜・*:.。..。.:*・'(*°▽°*)'・*:.。. .。.:*・゜ ゜・*			
Positive	*Like*	5	(*´v `)	(*б ω б*)	(^_-)	(*v ω v)	(*´ ω ` *)
Positive	*Relief*	9	(u_u)	(*´ɪ‘*)	d(´ ω ` *)	(´∀ ` *)	(´ __ ` 。)
			♪ (´▽ `)	(=´ ω ` =)	(o´▽ ` o)	(@￣ρ￣@)	
Negative	*Anger*	10	(` Д ´)	(* ` ω ´*)	(−_− #)	(‘皿´)	(# ` ∧ ´)
			(°Д°#)	o(` ω ´*)o	ヽ (` Д ´)ﾉ	(` ε ´)	(#°Д°)
Negative	*Dislike*	17	(; ￣ェ￣)	ヽ (´ ρ `)ﾉ	(´Д‘)	(− __− ;)	(´Д‘)
			(¬ з ¬)	(~_~;)	(´Д ` ;)	(￣ー￣)	(´Д ` υ)
			(-_-;)	(; ´Д `)	(; −_−)	(−_− ;)	(´A `)
			(￣(工)￣)	(ﾉД‘)			
Negative	*Fear*	1	(((((; ゜Д゜))))				
Negative	*Sadness*	27	(´△ `)	(; _ ;)/~~~	(つД `)	(; д ;)	(/□≦ 、)
			(T__T)	。・゜・(ﾉД‘)・゜・。	(´･_･‘)	(T_T)	(T . T)
			(T ^T)	(; __ ;)	。゜(゜´Д ` ゜)゜。	ヽ (; ▽ ;) ノ	(>_<)
			(´ ; ω ; `)	(ノД‘)	(/ _ ;)	(; ;)	(´･ω･‘)
			。゜゜(´□ ` 。)°゜。	。゜(。ﾉω ヽ。)゜。	(σ ´Д `) σ	。・°°・(>_<)・°°・。	
			il||li(つд-。)il||li	・゜・(つД `)・゜・	。・゜ ・(ノД‘)・゜ ・。		
Other	*Amazement*	22	(;゜д゜)	(; ゜ 0゜)	Σ (゜д゜|||)	(゜Д゜)	(・・?)
			(((゜Д゜)))	(゜д゜)	Σ (|||▽|||)	(O_O)	! (◎_◎;)
			Σ (゜Д゜)	(゜д゜)	(￣◇￣;)	Σ (・□・ ;)	(゜Д゜)
			(||゜Д゜)	(・・;)	(? _ ?)	Σ (||゜ Д゜)	(;￣O￣)
			Σ (´ д ` ;)	((((;゜Д゜))))))			
Other	*Excitement*	6	(>人<;)	(` ・д・ ´)	(*≧ д≦)	(` ・ω・ ´)	щ (゜Д゜щ)
			Ψ (` ∀ ´) Ψ				
Other	*Shyness*	10	(≧ 人≦)	(⌒-⌒;)	(/∀＼*)	(〃▽〃)	(^_^;)
			(^◇^;)	(〃ω〃)	(*ﾉɪﾉ)	(*ﾉωﾉ)	(。-_-。)

The reason our system does not recommend one emoticon but an ordered set of emoticons in Fig. 6 (2) was based on our hypothesis that there are high chances of different emoticons of the same emotion type being selected for the same context. For the prediction model, we employed machine learning with pre-trained models. We decided to test four different methods of creating our prediction model: (1) using Word2vec (Mikolov, Sutskever, Chen, Corrado, & Dean, 2013) vectors as features with SVM (Corinna & Vladimir, 1995), (2) using Word2vec embeddings with attention-based Bidirectional Long Short-Term Memory (BiLSTM) (Schuster & Paliwal, 1997; Yang et al., 2016), (3) using ELMo (Peters et al., 2018) embeddings with attention-based BiLSTM, and (4) fine-tuning with BERT language model (Devlin et al., 2019). The details of the pre-trained models are described in Section 4.1 and the four aforementioned prediction models are described in Section 4.2. Details of libraries, tools, and pre-trained models used for the implementation are described in Section 5.3.

4.1. Pre-trained models

4.1.1. Word2vec

Word2vec (Mikolov et al., 2013) is a two-layer neural network that reconstructs the linguistic contexts of words from natural language inputs. It learns the linguistic contexts using either of the two methods: continuous bag-of-words (CBOW) or continuous skip-gram (Skip-gram). In the CBOW model, the model predicts the current word by using a window of surrounding words. In the Skip-gram model, the distributed representation of the current word is used to predict the surrounding window of context words. By

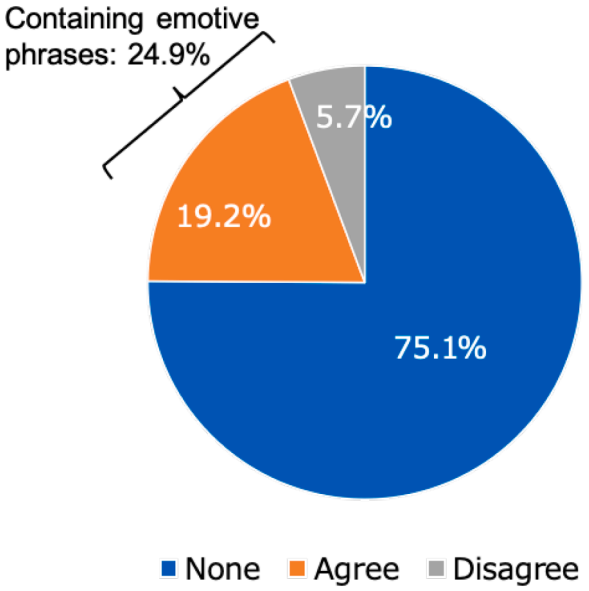
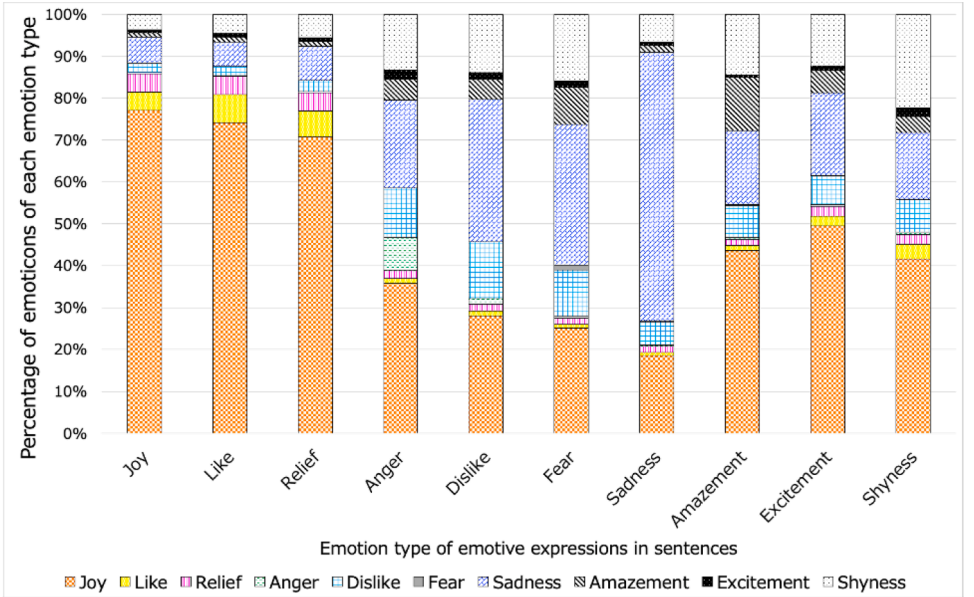


Fig. 4. Percentage of sentences containing emotive phrases. "Agree" indicates that polarity type of an emotive phrase matches emoticon's polarity type within a given sentence.



Number of sentences									
Joy	Like	Relief	Anger	Dislike	Fear	Sadness	Amazement	Excitement	Shyness
68,223 (10.54%)	34,527 (5.29%)	10,017 (1.53%)	1,598 (0.24%)	28,985 (4.44%)	4,576 (0.70%)	7,306 (1.12%)	2,743 (0.42%)	3,296 (0.50%)	1,054 (0.16%)

Fig. 5. Different ratios of emoticons' emotion types included in sentences categorized by emotion. The number of sentences for each emotion type is shown in the table and their percentage in brackets.

learning natural language texts with the Word2vec algorithm, we can retrieve distributed representation of words as vectors ("Embedding matrix" in Fig. 7) – and the words used in a similar context express nearly the same vectors. Fig. 7 is the Skip-gram model to explain the "Embedding matrix" which we used in the prediction model described in Sections 4.2.1 and 4.2.2. Both the input vector $(x_1, x_2, ..., x_V)$ and the output vector $(y_1, y_2, ..., y_V)$ are one-hot encoded-word representations. The hidden layer is the word embedding of size N. The "Embedding matrix" in the figure is the multiplication of the input vector X and the matrix W of size $V \times N$. V is the size of the vocabulary and N is the size of the embedding which can be set manually. Matrix W is determined by learning the semantic

Table 4
Examples of sentence-emoticon pairs that their emotion types do not match with each other. Emotive words are emphasized in bold.

Text	Emoticon	Text polarity	Text emotion	Emoticon polarity	Emoticon emotion
(1) アイコンさっき変えたんだけどみんな笑うの ¥¥(*I just changed my avatar but everyone **laughs about it***	(/_;)	POSITIVE	*Joy*	NEGATIVE	*Sadness*
(2) 恥ずかしさのあまり消去しました (*I deleted it because it was too **embarrassing***)	（；；）	OTHER	*Shyness*	NEGATIVE	*Sadness*
(3) 今日は思いっきり泣いてくださいね (*You can **cry** as much as you want today*)	(*^_^*)	NEGATIVE	*Sadness*	POSITIVE	*Joy*

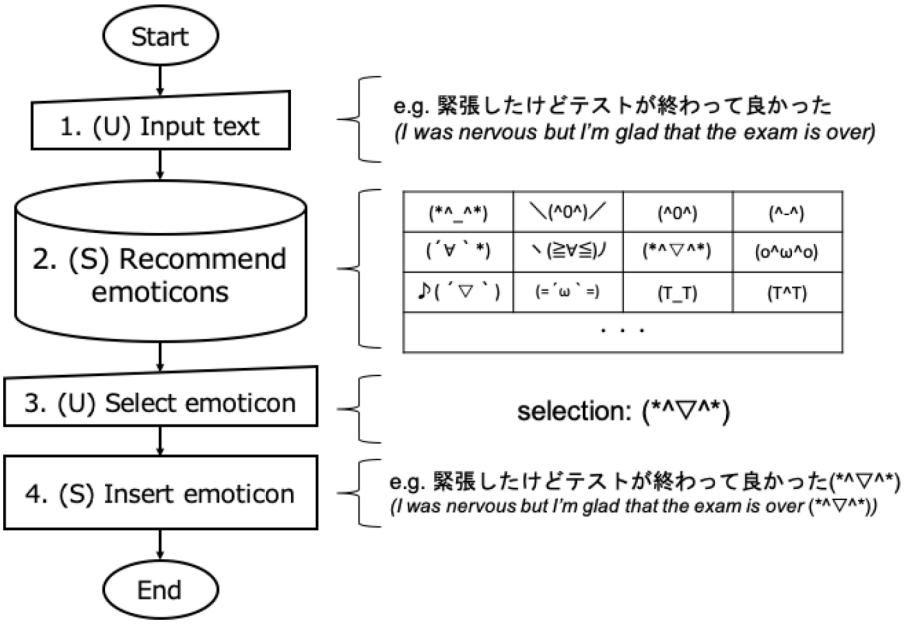


Fig. 6. System procedure.

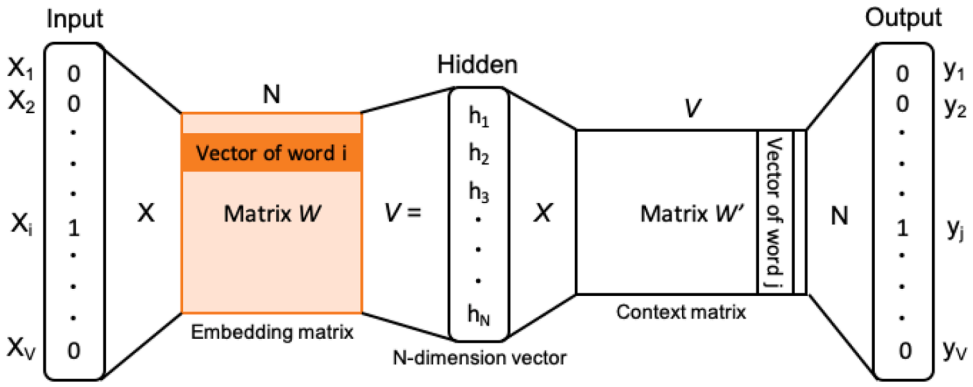


Fig. 7. Word2vec (Skip-gram) model. The figure was created with reference to (Weng, 2017).

relationships between words based on the surrounding words in a sentence. "Vector of word" in the figure means the embedding vector for the *i*th word in the vocabulary. "Context matrix" is the multiplication of the hidden vector and the matrix W' of size $N \times V$. Matrix W' is optimized to predict the surrounding words for the targeted input word. By using the "Embedding matrix", we can retrieve a static embedding vector for each word. The reason we decided to use Word2vec pre-trained model was our hypothesis that semantically similar texts tend to co-occur with similar emoticons and we considered that Word2vec vectors could grasp the semantic similarity of texts and predict more acceptable emoticons. We used vectors retrieved from Word2vec in SVM and attention-based BiLSTM, respectively (details are given in Sections 4.2.1 and 4.2.2).

4.1.2. ELMo

ELMo (Peters et al., 2018) allows us to obtain word representations that take the entire sentence or paragraph into account which is different from the Word2vec model which can only map a word to a fixed vector statically. Word embedding vectors retrieved from ELMo is a weighted sum of three layers (Fig. 8): word representation vectors that are created using a character-level convolutional neural network (CNN) (LeCun, Bengio, & Hinton, 2015) from a text string input, a hidden layer that is a concatenation of forward and backward information (BiLSTM) created from word representation vectors, and another hidden layer (BiLSTM) created from the previous hidden layer. In Fig. 8, s_0,s_1, and s_2 are weights for each layer. We employed a pre-trained ELMo model expecting to capture the context of an entire input sentence to recommend emoticons. We adapted features retrieved from the ELMo model with attention-based BiLSTM (details are given in Section 4.2.2).

4.1.3. BERT

BERT (Devlin et al., 2019) is a multi-layer bidirectional Transformer encoder that is trained on masked word prediction and next sentence prediction. Transformer (Vaswani et al., 2017) learns positions of words in a sentence and uses a multi-head attention mechanism to learn contextual relations between words in a text. As described in Devlin et al. (2019), BERT can learn the context of input sentences by learning both left and right context in all layers which enables the learning deeper than ELMo which only uses the concatenation of independently trained left-to-right and right-to-left LSTM. The fine-tuning approach using BERT has achieved remarkable results in many natural language understanding tasks. We applied a fine-tuning approach with a pre-trained BERT model in our emoticon recommendation system because we expected that this model could recommend emoticons regarding the contextual relations between words in an input text. The fine-tuning approach using BERT is described in Section 4.2.3.

4.2. Prediction models

4.2.1. Word2vec+SVM

Prediction model using vectors retrieved from the Word2vec model and SVM classifier (hereinafter referred to as *Word2vec+SVM*) is shown in Fig. 9. The following numbers correspond to the numbers in that figure.

1. Input sentence-emoticon pair as training data.
2. Divide sentences into tokens (tokenize).
3. Create feature vectors. The feature vector for each sentence was calculated with Eq. (1) using each word vector retrieved from the Word2vec model. In Eq. (1), assume that there are N words ($w_1, w_2, \ldots, w_N$) in a sentence s. Then, we retrieve a word embedding vector ($v_{w_i}, i = 1,2,\ldots,N$) of each word from Word2vec model and calculate the average of the word vectors (V_s in Eq. (1) becomes a feature vector of sentence s). We only used words that exist in the Word2vec model.

$$V_s = \frac{\Sigma_{i=1}^{N} v_{w_i}}{N} \tag{1}$$

4. Fit the training data to SVM model.

4.2.2. Word2vec/ELMo+BiLSTM+Attention

BiLSTM (Schuster & Paliwal, 1997) is one of the recurrent neural network methods that is composed of two recurrent network layers in which the output layer is concatenated with a layer processing the sequence forward and backward. This method is capable of learning the past and future information in the sequence. We also applied an attention mechanism (Yang et al., 2016) to learn important words in a sentence to recommend emoticons. The reason why we applied attention-based BiLSTM with Word2vec or ELMo models was that this combination has yielded good results in NLP tasks (e.g. emoji prediction and sentiment analysis) (Chen, Wang, & He, 2018; Peters et al., 2018). We also considered that the prediction result would be better if we not only predict emoticons by learning the sequence of words in texts but also if we focused on important words to predict emoticons. The description of the model using either pre-trained Word2vec model or ELMo model for word embeddings with attention-based BiLSTM (hereinafter referred to as *Word2vec+BiLSTM+Attention* and *ELMo+BiLSTM+Attention*) is shown in Fig. 10. The numbers below correspond to the numbers in the figure.

1. Input layer: input sentences to the model (sentences are tokenized).
2. Embedding layer: convert the input layer to a word embedding vector retrieved from Word2vec or ELMo model. The concrete way of retrieving a word embedding vector from each Word2vec and ELMo differs from each other: (Word2vec) input each word and retrieve static word embedding vector from the pre-trained model. (ELMo) input the entire sentence and retrieve a weighted sum of the three layers (see Section 4.1.2) from the pre-trained model. For the matrix size, for instance, if there are 20 tokens in a sentence and the dimension of a pre-trained model is 200, the matrix size of the embedding layer will be 20×200.
3. LSTM layer: input embedding layer into BiLSTM layer. BiLSTM layer consists of forward hidden state $\overrightarrow{h_i} \in \mathbb{R}^{d_h}$ and backward hidden state $\overleftarrow{h_i} \in \mathbb{R}^{d_h}$, where d_h denotes the number of the hidden units. We concatenated both layers (i.e. $h_i = [\overrightarrow{h_i}; \overleftarrow{h_i}] \in \mathbb{R}^{2d_h}$).

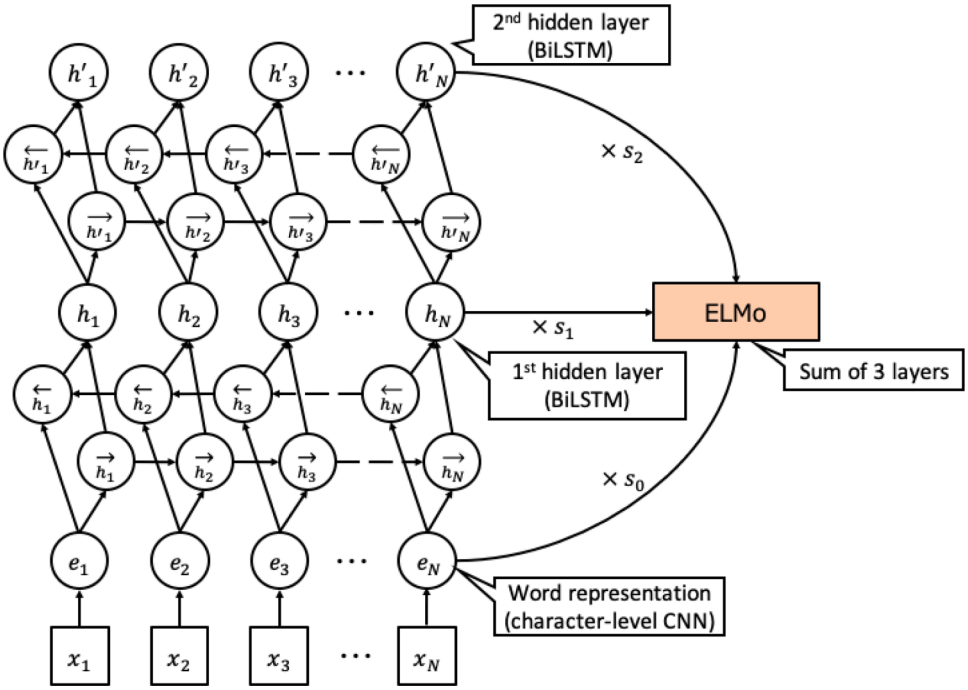


Fig. 8. ELMo model.

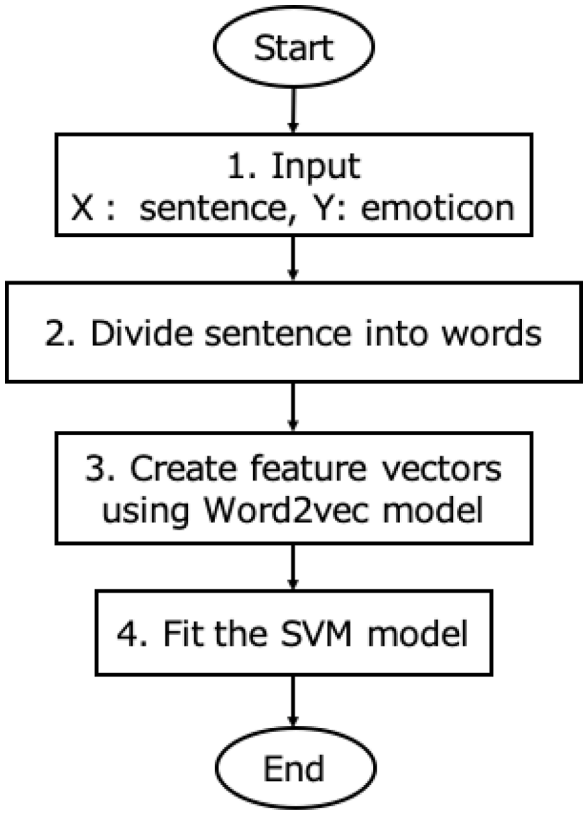


Fig. 9. *Word2vec+SVM* model.

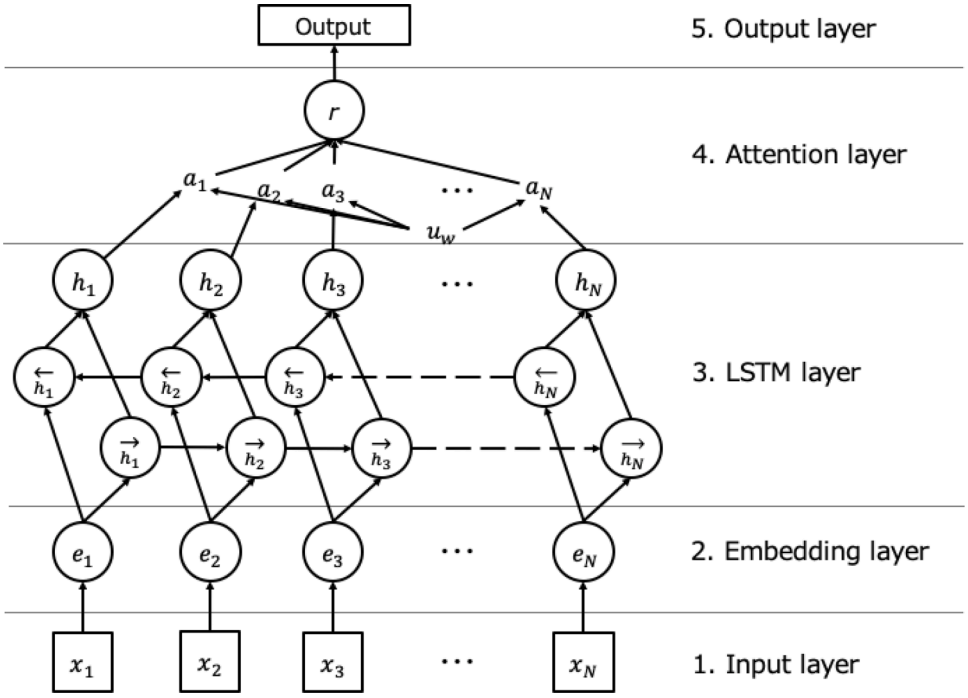


Fig. 10. *Word2vec/ELMo+BiLSTM+Attention* model.

4. Attention layer: feed the LSTM layer into the Attention layer. We implemented an attention method described in Yang et al. (2016) which is also used in other NLP classification tasks and has shown its effectiveness (Li, Rzepka, Ptaszynski, & Araki, 2019; Li, Li, Hu, Zhang, & Hu, 2018; Yang et al., 2016). The attention layer calculates the representation r based on the three equations written below (Eqs. (2)–(4)). We first feed the hidden state $h_i$through a one-layer Multi-Layer Perceptron to get hidden representation

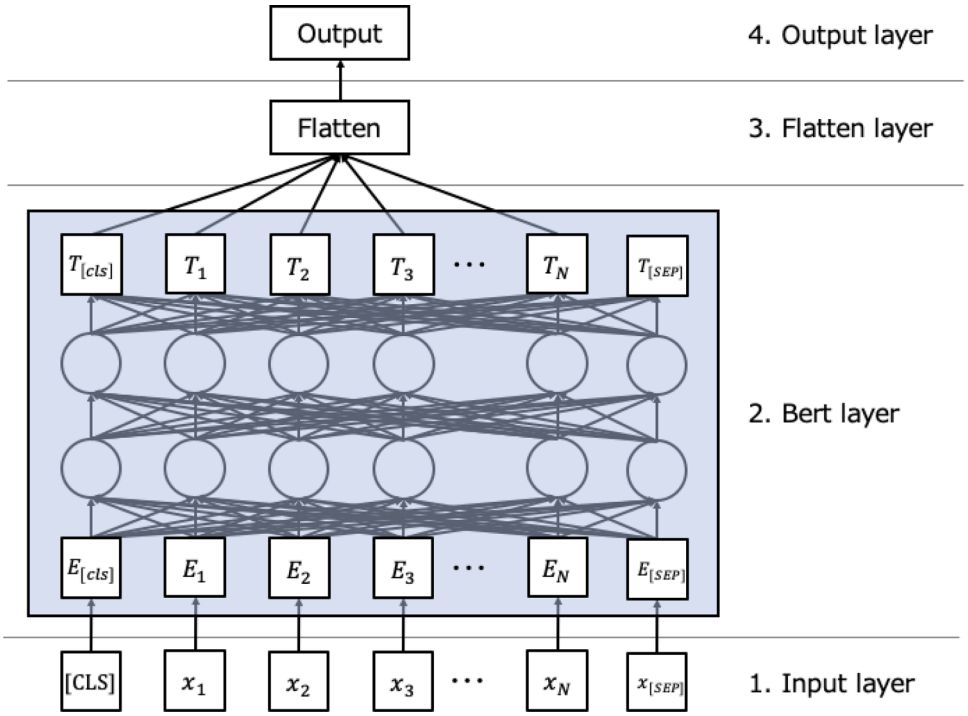


Fig. 11. *BERT+fine-tuning* model.

u_i(Eq. (2)). W_w and b_w are weight and bias, respectively, and they are optimized during the training period. The weight value $a_i$for $h_i$is computed as the similarity between u_i and a word-level context vector u_w(Eq. (3)). The calculation for weight value uses softmax function to convert the value into a normalized probability distribution. Finally, the representation r is computed as a weighted sum of the hidden state vectors with weights (Eq. (4)). The context vector u_w is considered as a high-level representation to judge the importance of different words in a sentence (Yang et al., 2016). It can be considered as a "weight" used with the hidden representation u_i and is randomly initialized to allow different weights to learn independently of each unit. It is also jointly learned during the training process.

$$u_i = tanh(W_w h_i + b_w) \tag{2}$$

$$a_i = softmax(u_i^T u_w) = \frac{exp(u_i^T u_w)}{\Sigma_{i=1}^{N} exp(u_i^T u_w)} \tag{3}$$

$$r = \Sigma_{i=1}^{N} a_i h_i \tag{4}$$

5. Output layer: the output is decoded to a probability of each emoticon using softmax function.

4.2.3. *BERT+fine-tuning*

The description of the fine-tuning approach using BERT (hereinafter referred to as *BERT+fine-tuning*) is given below. The numbers below correspond to the numbers in Fig. 11.

1. Input layer: input a sentence into the model. Sentences are tokenized and "[CLS]" is added before the first token and "[SEP]" is added after the last token as explained in Devlin et al. (2019).
2. BERT layer: input the input layer into the BERT layer. As described in Devlin et al. (2019), the input embeddings ($E_{[cls]}, E_1, ..., E_{[sep]}$) are the sum of the token embeddings, the segmentation embeddings, and the position embeddings (Fig. 12). The matrix size of the BERT layer output is the number of tokens in a sentence, each having 768 dimensions. For example, when the number of tokens used for learning is 20, then the matrix size of the output is 20×768. The number 768 is a default number of dimensions used in Devlin et al. (2019).
3. Flatten layer: input the BERT layer into the Flatten layer to convert the matrix into a single array. For example, if the Flatten layer receives a matrix size of 20×768described in the previous step, it transforms into a single array containing 15,360 ($= 20 \times 768$) dimensions.
4. Output layer: the output is decoded to a probability of each emoticon using softmax function.

5. Evaluation

We investigated which prediction model recommends desired emoticons best by performing two evaluations: (A) recommendation accuracy and (B) Mean Average Precision (MAP) (Steck, 2013). MAP is a method common in the recommendation technology field which evaluates how many of the relevant items are recommended as the top candidates from the entire pool of recommended items. (A) Recommendation accuracy is based on the number of correct emoticons recommended within the top 25. The reason we set the number to 25 was that the number of emoticons shown on the smartphone as a default is about 25 emoticons (the number changes depending on the length of emoticons). Thus, we considered that the prediction model is effective if the desired emoticon is recommended among the first 25 emoticons which users see on their screens at first glance (without scrolling down). By "correct emoticons" we mean all facemarks used in sentences of the test data (details of the test data are described in Section 5.1). We evaluated (B) MAP of each prediction model to compare which one is likely to recommend emoticons of the same polarity or the emotion type of the correct emoticon at the top of the list. As users tend to choose emoticons of the same polarity/emotion type for the similar context, we considered that the prediction model is effective if the emoticons of the same polarity/emotion are recommended with the correct emoticon at top of the list. We evaluated how emoticons of the same polarity are recommended at top of the list with the correct

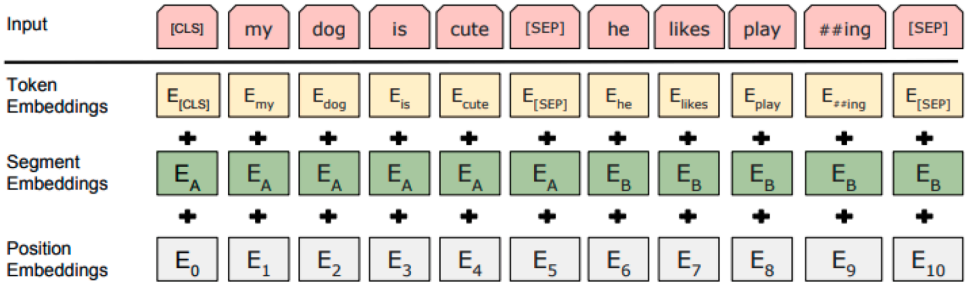


Fig. 12. BERT input representation, borrowed from (Devlin et al., 2019).

emoticon (hereinafter referred to as "MAP (polarity)") and emoticons of the same emotion are recommended at top of the list with the correct emoticon (hereinafter referred to as "MAP (emotion)"). We excluded the results for OTHER, namely (*Amazement, Excitement*, and *Shyness*) type emoticons when calculating MAP (polarity) because we considered *Amazement, Excitement*, and *Shyness* as not similar to each other (also they can be either positive or negative).

Recommendation accuracy was calculated by using Eq. (5). *A@N* indicates the accuracy of correct emoticons recommended within the top *N*. For example, *A*@5 means the percentage of correct emoticons recommended was within the top 5 output candidates. *S*denotes the number of samples and r_i is the ranking of the correct emoticon for *i*-th sample. MAP was calculated by using Eqs. (6) and (7). In Eq. (6), *S*denotes the number of data samples. AP of a sample *s* is calculated with Eq. (7). In Eq. (7), *M*denotes the number of desired emoticons for the sample and $Precision@k \cdot y_k$ indicates the precision of relevant items within top *k*. $Precision@k$ is calculated with Eq. (8). $y_k$is set to 1 when the desired emoticon is ranked at k and if not y_k is set to 0. For example, suppose that the relevant items are in rank 1 and 3. AP is then $\frac{1}{2} \times \left(\frac{1}{1} + \frac{2}{3}\right) = 0.83(pt)$. The MAP value close to 1 means the prediction model predicts relevant emoticons at the top of the list and a value close to 0 means that relevant emoticons are recommended at the bottom of the list.

$$A@N = \frac{\Sigma_{i=1}^{S} f(r_i)}{S} \times 100(\%), f(r_i) = \begin{cases} 1, & \text{if } r_i \leqq N \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

$$MAP = \frac{1}{S}\Sigma_{s \in S} AP(s) \tag{6}$$

$$AP(s) = \Sigma_{k=1}^{M} \frac{Precision@k \cdot y_k}{\Sigma_{i=1}^{k} y_i}, y_k = \begin{cases} 1, & \text{emoticon ranked at } k \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

$$Precision@k = \frac{\text{Number of desired emoticons within top k}}{k} \tag{8}$$

We compared the proposed prediction models to conventional methods described in Section 5.2. As described in Bernico, Li, and Zhang (2018), the domain similarity of source and target tasks is important to achieve higher prediction results, therefore, we also investigated whether the domains of pre-trained models influenced emoticon prediction by comparing multiple pre-trained models. Details of parameters and libraries used for each method are described in Section 5.2 for baseline methods and in Section 5.3 for proposed methods.

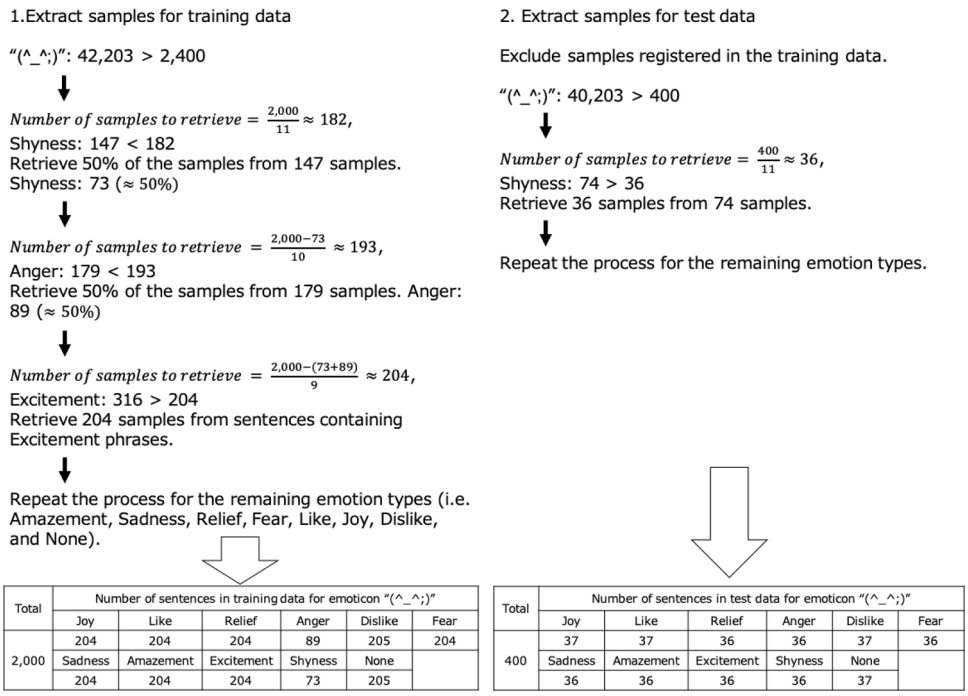
5.1. Data description

The data described in Section 3 was imbalanced in terms of the number of sentences containing each emoticon. Therefore, we collected a maximum of 2000 data samples for each emoticon for training data and a maximum of 400 data samples for each emoticon for the test data. We collected the samples of each emoticon based on a procedure described in Fig. 13 to collect sentences containing emotive statement for each of 10 emotion types and NONE sentences evenly. We collected samples for training and test data in this way because the whole data contained 75% of sentences not including emotive phrases and may result in collecting imbalanced data if we collected samples randomly.

As shown in Fig. 13, there are 42,203 samples for emoticon "(^_^;)" in the data described in Section 3. If we collect the samples randomly, we may collect more samples categorized as NONEand less samples from the other emotion types because 80% of the samples are included in the NONE type. To tackle this problem, we first collected samples for the training data as follows. First, we checked whether the number of samples is more than 2400 (the biggest set of samples for an emoticon in training and test data). If the number of samples is more than 2400, then we computed the number of samples to retrieve by dividing 2000 with the number of existing emotion types for the emoticon including NONE. For example, in Fig. 13, samples exist for all emotion types, therefore, the initial number of samples to retrieve is 182 ($= 2000/11$). Then, we retrieved samples from the emotion type that contained the smallest number of samples for the emoticon (in Fig. 13, *Shyness*, 147 samples). If the number of samples is smaller than the number we need to collect, we only collected 50% of the samples so we can keep the rest in the test data. In Fig. 13, we collected 73 sentences for *Shyness* because the number of samples was 147 which was less than 182. Next, we recalculated the average number of samples to collect by dividing the remaining total number we need to collect with the rest of the emotion types. For example, the number of samples to retrieve changed to 193, dividing 1927 samples ($= 2000 - 73$) with the rest of the 10 emotion types. Then, we collected samples from the second smallest emotion type (for example, *Anger* in Fig. 13). We repeated the calculation to determine the number of samples to retrieve for each emotion type and retrieved samples for training data. The procedure to retrieve samples for test data was the same from training data but the maximum number of retrieval was 400 samples. If the number of samples in the data (Section 3) for the certain emoticon is less than 2400, we collected 50% of samples for each emotion type for training data and the remaining samples were registered in test data.

The size of the training data was 124,359 samples containing 167 different emoticons. The number of samples that contained an emotive word of a certain emotion type and the number of samples that do not contain emotive words are shown in Table 5. The size of the test data was 37,622 samples containing the same number of different emoticons in the training data. The number of samples containing an emotive word of a certain emotion type and the number of samples without any emotive words are shown in Table 6.

Total	Number of sentences containing emotional phrases of emotion type for emoticon "(^_^;)"					
	Joy	Like	Relief	Anger	Dislike	Fear
42,203	1,769	848	423	179	3,434	615
	Sadness	Amazement	Excitement	Shyness	None	
	382	332	316	147	33,758	



Total	Number of sentences in training data for emoticon "(^_^;)"					
	Joy	Like	Relief	Anger	Dislike	Fear
2,000	204	204	204	89	205	204
	Sadness	Amazement	Excitement	Shyness	None	
	204	204	204	73	205	

Total	Number of sentences in test data for emoticon "(^_^;)"					
	Joy	Like	Relief	Anger	Dislike	Fear
400	37	37	36	36	37	36
	Sadness	Amazement	Excitement	Shyness	None	
	36	36	36	36	37	

Fig. 13. Procedure for collecting training/test data samples of each emoticon.

Table 5
Samples for each emotion type (training data).

	Joy	*Like*	*Relief*	*Anger*	*Dislike*	*Fear*	*Sadness*	*Amazement*	*Excitement*	*Shyness*	NONE
Sample	11,600	7935	4464	822	9046	2196	2658	1470	1,730	566	81,872
Total	124,359										

Table 6
Samples for each emotion type (test data).

	Joy	*Like*	*Relief*	*Anger*	*Dislike*	*Fear*	*Sadness*	*Amazement*	*Excitement*	*Shyness*	NONE
Sample	3867	2696	1450	639	3729	1319	1424	901	1043	480	20,074
Total	37,622										

5.2. Baseline methods

We compared whether applying pre-trained models was more effective than conventional methods:

- *Frequency*: recommend emoticons based on emoticon frequency in the training data
- *BoW+SVM*: Bag of Words (BoW) as features with SVM classifier
- *TFIDF+SVM*: term frequency inversed document frequency (TFIDF) as features with SVM classifier
- *WordCharGram+SVM*: n-gram of words and characters as features with SVM classifier (Çöltekin & Rama, 2018)

We employed *Frequency*, which we believe it to be a similar algorithm to Emura et al.'s method (Emura & Seki, 2012), to compare our method to the previously proposed rule-based method. *WordCharGram+SVM* was employed because this method worked best in SemEval emoji prediction task (Barbieri et al., 2018). *BoW+SVM* and *TFIDF+SVM* were employed because these methods are widely used in various NLP tasks as baselines. For *Frequency*, we recommended frequent emoticons based on the emotion type of emotive

expressions if it exists in an input. For example, if an emotive expression exists in a sentence and it expresses *Joy*, we recommended frequent emoticons from training data that contained *Joy* type emotive expressions in sentences. If no such expressions existed in an input, the system recommended frequent emoticons from the training data that do not contain emotive expressions in sentences. For *WordCharGram+SVM*, we used bigrams of words and characters as features, respectively, as it achieved the best A@25 among 2–5 grams. For SVM in *BoW+SVM, TFIDF+SVM*, and *WordCharGram+SVM*, we implemented these models using Scikit-learn (Pedregosa et al., 2011) library. We employed C-Support Vector Classification (C-SVC) with Radial Basis Function (RBF) kernel which is the default kernel. We determined hyperparameters, gamma (γ), and C by performing grid search ranging from 0.01 to 10 and used the parameters that yielded the best A@25 on the test data. The gamma parameter defines how strong each training sample influences the prediction and the C parameter is a trade-off of correct classification of training examples against maximization of the decision function's margin.[7] The best parameters for *BoW+SVM, TFIDF+SVM, WordCharGram+SVM* are listed in Table 7. Our system sorted the emoticons in order based on the value of decision function which gives the probability of the prediction of each emoticon. Since the text data was in Japanese in which words are not divided by spaces, we used a morpheme analysis tool called MeCab (Kudo, Yamamoto, & Matsumoto, 2004) for tokenization.

5.3. Implementation for each proposed prediction model

5.3.1. Word2vec+SVM

We used Gensim (Řehůřek & Sojka, 2010) library to load the pre-trained Word2vec model and Scikit-learn library to implement SVM. We utilized two pre-trained Word2vec models to create features from texts: *Word2vec (Wiki)*, a 300-dimensional model learned from Japanese Wikipedia articles[8] and *Word2vec (SNS)*, a 200-dimensional model learned from social media corpus[9]. These two pre-trained models were used separately to examine whether the difference in the learning domain (Wikipedia or social media) influences the recommendation results. Since emoticons are more likely to be used in casual contexts, we expect that using a pre-trained model trained on social media corpus will recommend better results. We used MeCab for tokenization because this tool was also used to create pre-trained Word2vec models. Words that do not exist in Word2vec were excluded. 87.8% and 78.7% of all words in both training and test data existed in Word2vec (SNS) and Word2vec (Wiki), respectively. We used the parameters of gamma and C that yielded the best A@25 with test data. For the model using *Word2vec (Wiki)*, gamma and C were 1.0 and 10, respectively (see Table 8, *Word2vec (Wiki)+SVM*). For the model using *Word2vec (SNS)*, gamma and C were 0.01 and 10, respectively (see Table 8, *Word2vec (SNS)+SVM*). The system sorted the emoticons in order based on the value of decision function which gives the probability of the prediction for each emoticon.

5.3.2. Word2vec+BiLSTM+Attention

We employed Tensorflow Keras[10] library to create deep learning-based model. Tensorflow Keras was also used for implementing *ELMo+BiLSTM+Attention* and *BERT+fine-tuning*. We used MeCab to tokenize texts for the same reason explained in Section 5.3.1. As deep learning methods using sequential data require fixed length, we used the last 20 tokens for each sentence when feeding into the input layer. We set the token number to 20 because 94.2% of the sentences contained less than 20 tokens. We did not use the first 20 tokens but the last 20 tokens because the latter words have more influence to choose emoticons (Emura & Seki, 2012). For the embedding layer, we used the same Word2vec models described in Section 5.3.1. For the BiLSTM layer, we set 200 units (100 units for each forward and backward processing) and the batch size was set to 16. We set the learning rate among 0.01 to 1e-6 by investigating which learning rate yields the best A@25 for the test data. The best learning rate was 1e4 (see Table 9, *Word2vec (Wiki)+BiLSTM+Attention*and *Word2vec (SNS)+BiLSTM+Attention*). We ran the learning until 20 epochs and the number of epochs that yielded the best A@25 for the test data was 13 epochs for *Word2vec (Wiki)+BiLSTM+Attention* and 8 epochs for *Word2vec (SNS)+BiLSTM+Attention*. We applied Adam (Kingma & Ba, 2015) as an optimizer which is commonly used and known as effective in natural language-based tasks. We used categorical cross-entropy as the loss function which is calculated by Eq. (9). In Eq. (9), $y$is an indicator giving either 0 or 1 indicating whether a class label is correctly classified, and $P$is the predicted probability of the correctly classified label. $i \in [1,S]$is the index of training samples, and $j \in [1,N]$is the index of the class label. Adam and categorical cross-entropy are also used in *ELMo+BiLSTM+Attention* and *BERT+fine-tuning*.

$$Loss = -\Sigma_{i=1}^{S}\Sigma_{j=1}^{N}\left(y_{ij}log\left(P_{ij}\right)\right) \tag{9}$$

5.3.3. ELMo+BiLSTM+Attention

We used ELMoForManyLangs[11] library to retrieve word vectors as embedding layer from pre-trained ELMo model. We employed a

[7] For details, see https://scikit-learn.org/stable/auto_examples/svm/plot_rbf_parameters.html

[8] Word2vec model trained on Wikipedia Japanese articles: https://github.com/singletongue/WikiEntVec/releases/jawiki.all_vectors.300d.txt.bz2, created by M. Suzuki.

[9] Word distributed representation model from the largest Japanese social media corpus: https://github.com/hottolink/hottoSNS-w2v, created by a Japanese company, HottoLink.

[10] Tensorflow-keras: https://www.tensorflow.org/guide/keras, version 2.1.0-rc2.

[11] ELMoForManyLangs: https://github.com/HIT-SCIR/ELMoForManyLangs, version 0.0.3.

Table 7
The best parameters for each baseline method.

Method	γ(SVM)	C (SVM)
BoW+SVM	0.1	10
TFIDF+SVM	0.1	10
WordCharGram+SVM	0.1	10

Table 8
The best parameters for each prediction model (models using SVM) and number of dimensions of pre-trained models.

Method	γ(SVM)	C (SVM)	Dimension
Word2vec (Wiki)+SVM	1.0	10	300
Word2vec (SNS)+SVM	0.01	10	200

Table 9
The best parameters for each prediction model (models using deep learning techniques).

Method	Dimension	LSTM unit	Batch size	Epoch	Learning rate
Word2vec (Wiki)+BiLSTM+Attention	300	200	16	13	$1e-4$
Word2vec (SNS)+BiLSTM+Attention	200	200	16	8	$1e-4$
ELMo+BiLSTM+Attention	1024	200	16	4	$1e-5$
BERT (Wiki)+fine-tuning	768	-	16	4	$1e-5$
BERT (SNS)+fine-tuning	768	-	16	18	$1e-6$

pre-trained ELMo model[12] which was trained on 400,000 Japanese business news articles as it was the only Japanese ELMo model freely available. We used MeCab for tokenization to match the text chunks in the pre-trained ELMo model. For the input layer, we used the last 20 tokens for each input sentence which is as same as in Section 5.3.2. We set 200 units for the BiLSTM layer and the batch size was set to 16. The learning rate and the number of epochs for the model we used for the comparison were 1*e*-5 and 4 epochs, respectively (see Table 9, *ELMo+BiLSTM+Attention*).

5.3.4. *BERT+fine-tuning*

We used Bert-for-tf2[13] to load the pre-trained BERT model. We employed two pre-trained BERT models: *BERT (Wiki)*, a model trained on Japanese Wikipedia articles (Kikuta, 2019) and *BERT (SNS)*, a model trained on social media corpus (Sakaki, Mizuki, & Gunji, 2019). SentencePiece (Kudo & Richardson, 2018) was utilized for tokenization. For the input layer, we used the last 20 tokens for each input sentence which is as same as in Sections 5.3.2 and 5.3.3. The learning rate and the number of epochs for the model we used for the comparison were 1*e*5 and 4 epochs, respectively, for the *BERT+fine-tuning*model using pre-trained *BERT (Wiki)* model (see Table 9, *BERT (Wiki)+fine-tuning*). The learning rate and the number of epochs for the model we used for the comparison were 1*e*6 and 18 epochs, respectively, for BERT+fine-tuning model using *BERT (SNS)*model (see Table 9, *BERT (SNS)+fine-tuning*). The batch size was set to 16.

6. Results and discussions

6.1. *Recommendation accuracy*

Table 10 shows the recommendation accuracy of correct emoticons recommended within the top 25. We also divided the test data according to the emotion type of emotive expression in a sentence to see whether there is a difference in the accuracy between sentences that contain an emotive expression of a certain emotion category including None type sentences. We performed Wilcoxon signed-rank test (Wilcoxon, 1992) to investigate a statistical difference between rankings of the correct emoticon for each sample given by the prediction models (see Table 11). Wilcoxon signed-rank test is a non-parametric statistical test to compare two related samples to assess whether the population mean ranks differ. The *P*-value lower than 0.05 commonly indicates that the two samples are significantly different from each other. The *P*-value of all pairs except for one pair, *BoW+SVM* & *TFIDF+SVM* were lower than 0.01 indicating that there are significant differences in the recommendation results outputted by the prediction models. The P-value for the pair *BoW+SVM* & *TFIDF+SVM* was 0.06.

We confirmed that *BERT (SNS)+fine-tuning* showed the highest accuracy of 52.98% and *Frequency* showed the lowest accuracy of 29.53% among all the prediction models for the entire test data (see Table 10). We were able to confirm that models using pre-trained

[12] ELMo model: https://stockmark.co.jp/news/2019/01/25/783/, created by a Japanese company, Stockmark Inc.
[13] Bert-for-tf2: https://github.com/kpe/bert-for-tf2/, version 0.13.5.

Table 10
Recommendation accuracy (A@25) (%). (1) *Frequency*, (2) *BoW+SVM*, (3) *TFIDF-SVM*, (4) *WordCharGram+SVM*, (5) *Word2vec (Wiki)+SVM*, (6) *Word2vec (SNS)+SVM*, (7) *Word2vec (Wiki)+BiLSTM+Attention*, (8) *Word2vec (SNS)+BiLSTM+Attention*, (9) *ELMo+BiLSTM+Attention*, (10) *BERT (Wiki)+fine-tuning*, (11) *BERT (SNS)+fine-tuning*.

		Emotion type of emotive expression in inputs (with number of samples)										
Method	Average A@25	*Joy*1930	*Like* 1494	*Relief* 942	*Anger* 420	*Dislike* 2047	*Fear* 812	*Sadness* 947	*Amazement* 588	*Excitement* 720	*Shyness* 324	*NONE*9708
(1)	29.53	24.44	32.53	52.69	**68.23**	32.23	56.41	60.53	61.38	59.73	**70.62**	18.75
(2)	42.37	40.11	45.51	56.83	56.34	45.53	55.12	54.56	55.27	57.14	60.42	36.83
(3)	42.48	41.09	46.81	57.03	55.71	45.19	55.50	54.92	56.94	59.06	60.00	36.52
(4)	43.58	43.55	49.59	59.93	55.87	46.63	55.80	54.00	58.38	59.44	60.83	37.20
(5)	44.18	43.37	49.33	57.79	56.65	45.83	53.37	55.69	58.27	59.25	57.50	38.81
(6)	45.62	44.45	50.63	60.00	59.31	48.62	56.33	57.51	58.82	60.02	58.54	39.93
(7)	47.63	43.75	51.56	62.00	59.15	50.04	60.73	63.06	62.38	63.57	65.62	42.11
(8)	48.37	44.63	49.07	60.83	62.44	50.01	60.27	64.54	63.26	65.20	66.46	43.45
(9)	47.42	44.48	50.30	58.97	59.62	48.78	57.70	60.67	62.93	61.74	58.33	42.81
(10)	50.43	47.30	51.48	62.48	61.97	50.15	**62.47**	64.19	63.60	64.72	65.62	46.24
(11)	**52.98**	**49.91**	**55.42**	**64.97**	65.73	**54.89**	61.56	**66.50**	**65.26**	**69.03**	67.50	**48.36**

Table 11
P-value of Wilcoxson signed-rank test between each method. Only one result was not statistically significant; it is marked with "†". (1) *Frequency*, (2) *BoW + SVM*, (3) *TFIDF + SVM*, (4) *WordCharGram + SVM*, (5) *Word2vec (Wiki) + SVM*, (6) *Word2vec (SNS) + SVM*, (7) *Word2vec (Wiki) + BiLSTM + Attention*, (8) *Word2vec (SNS) + BiLSTM + Attention*, (9) *ELMo + BiLSTM + Attention*, (10) *BERT (Wiki) + fine-tuning*, (11) *BERT (SNS) + fine-tuning*.

Method	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(1)	–										
(2)	0.00	–									
(3)	0.00	0.06†	–								
(4)	0.00	2.47e − 14	1.72e − 12	–							
(5)	0.00	2.21e − 46	1.93e − 44	1.37e − 22	–						
(6)	0.00	1.18e − 109	4.07e − 107	2.08e − 71	1.29e − 23	–					
(7)	0.00	0.00	0.00	0.00	2.04e − 269	1.90e − 144	–				
(8)	0.00	0.00	0.00	0.00	1.31e − 283	1.46e − 181	1.08e − 8	–			
(9)	0.00	7.45e − 299	1.56e − 296	2.83e − 271	2.24e − 142	2.39e − 71	2.59e − 5	4.93e − 20	–		
(10)	0.00	0.00	0.00	0.00	3.08e − 267	6.66e − 172	6.50e − 22	2.44e − 10	2.12e − 41	–	
(11)	0.00	0.00	0.00	0.00	0.00	0.00	5.85e − 225	1.57e − 183	1.10e − 237	1.22e − 86	–

Word2vec, ELMo, and BERT outperformed conventional methods by 0.60 pt (difference of 44.18% for *Word2vec (Wiki)+SVM* and 43.58% for *WordCharGram+SVM*) to 23.45 pt (difference of 52.98% for *BERT (SNS)+fine-tuning* and 29.53% for *Frequency*). For sentences not containing emotive phrases, our proposed methods successfully achieved higher recommendation accuracy than the conventional methods by 1.61 pt (difference of 38.81% for *Word2vec (Wiki)+SVM* and 37.20% for *WordCharGram+SVM*) to 29.61 pt (difference of 48.36% for *BERT (SNS)+fine-tuning* and 18.75% for *Frequency*). For sentences that contain emotive expressions, we confirmed that our proposed method achieved higher accuracy than the conventional methods for eight emotion types except *Anger* and *Shyness*.

We further investigated the reason why the results of *Frequency* scored the highest for inputs that contained *Anger* and *Shyness* expressions. From our analysis, we concluded two reasons. The first reason was that sentences containing expressions of *Anger* or *Shyness* co-occurred with fewer number of unique emoticons (see Fig. 14, striped bars). Moreover, the emoticons for around 70% of sentences in these data were concentrated in the top 25 frequently used emoticons in the training data which is higher than the sentences containing expressions of other emotion types or sentences without emotive expressions. The second reason was that some similar sentences containing expressions of *Anger* or *Shyness* co-occurred with emoticons of different emotion types which made the prediction result less adequate for the machine learning-based models. For example, as shown in Table 12, we collected inputs that contained a word ``怒る" (*okoru*, "angry"). We can see that emoticons expressing *Anger, Sadness*, and *Shyness* are present in the training

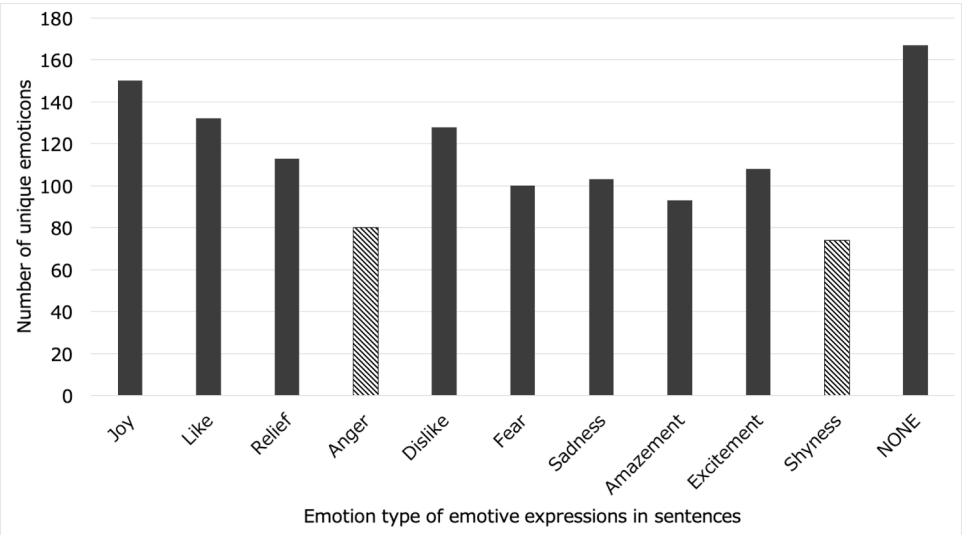


Fig. 14. Number of unique emoticons that appeared with the emotive expression in sentences in the training data. The top two lowest bars (*Anger* and *Shyness*) are emphasized with stripes to mirror the discussion written in Section 6.1.

Table 12
Examples of sentences containing emotive expression, 怒る (*okoru*, "angry").

Data	Text	Emoticon	Emotion type
Training	絶対怒ってるよね (*He/She must be* ***angry***.)	(`ε´)	*Anger*
Training	ドクター怒っちゃうのね (*The doctor is getting* ***angry***.)	(T_T)	*Sadness*
Training	顔が怒ってらっしゃる (*The face is filled with* ***anger***.)	(¯;)	*Shyness*
Test	めっちゃ怒ってるやーん (*You're so* ***angry***.)	(⌒-⌒;)	*Shyness*
Test	顔が怒ってるよ (*Your face seems* ***angry***.)	(;゜0゜)	*Amazement*
Test	怒ってるように見えました (*The person looked* ***angry***.)	(ᵔ)	*Joy*

data while emoticons expressing *Shyness, Amazement*, and *Joy* exist in the test data. For the reference, the number of sentences that contained the word ``怒る" (*okoru*, "angry") in the training data was 357, and this word co-occurred with emoticons mostly expressing *Joy* (30.5%), *Shyness* (27.7%), and *Sadness* (23.5%). The number of inputs that contained this word in the test data was 259, and it co-occurred with emoticons mostly expressing *Joy* (32.8%), *Sadness* (25.5%), and *Dislike* (13.1%) showing that the distribution of used emoticons is slightly different from the training data. To provide another example (shown in Table 13), we collected inputs which contain a word ``恥ずかしい" (*hazukashii*, "embarrassed"). We can also observe here that emoticons expressing *Dislike* and *Shyness* are found in the training data while emoticons expressing *Sadness* and *Shyness* in the test data. For the reference, the number of sentences that contain the word ``恥ずかしい" (*hazukashii*, "embarrassing") in the training data was 211, and it co-occurred with emoticons mostly expressing *Shyness* (43.6%), *Joy* (29.4%), *Sadness* (11.8%), and *Dislike* (9.5%). The number of inputs that contained this word in the test data was 134, and it co-occurred with emoticons mostly expressing *Joy* (39.6%), *Sadness* (21.6%), and *Shyness* (14.5%), showing again that the distribution of used emoticons is different from the training data. We assume that these two reasons occurring together led to the higher results for *Frequency* when compared to other machine learning methods for inputs containing *Anger* or *Shyness* expressions.

6.2. MAP

The results of MAP evaluation are shown in Table 14. We also divided the test data according to the emotion types of emotive expression in sentences to see whether there is a difference in the accuracy between sentences containing an emotive expression of a certain emotion type. Overall, *BERT (SNS)+fine-tuning* achieved the best results of 0.54 for polarity and 0.32 for emotion categorization. This indicates that emoticons agreeing on polarity/emotion type with the correct emoticon are recommended at the top of the list especially when using this prediction model. We did not confirm different tendencies for both MAP (polarity) and MAP (emotion) among sentences containing different emotion types of emotive expressions and NONE type sentences. We experimentally confirmed that all our proposed methods (5)–(11) outperformed the conventional methods (1)–(4). *BERT+fine-tuning* (See (10) and (11) in Table 14) achieved the highest scores for both sentences containing an emotive expression of each emotion type and for sentences without any emotive expression.

Figs. 15 and 16 show the distribution of average precision (AP) for polarity and emotion aspects of all samples in the test data. In these figures, the AP close to 1 means the prediction model predicts relevant emoticons at the top of the list and the value close to 0 means that relevant emoticons are recommended at the bottom of the list. From these figures, we confirmed that AP (polarity) was mostly distributed between 0.20 to 0.60 and 0 to 0.40 for AP (emotion) for samples predicted by *Frequency* (see (1) in Figs. 15 and 16). AP of other conventional prediction methods, *Bow + SVM, TFIDF + SVM*, and *WordCharGram +SVM* and the proposed methods *Word2vec(Wiki/SNS)+SVM*, was mostly distributed around 0.40 to 0.60 for polarity and 0 to 0.60 for emotion types (see (2)–(6) in Figs. 15 and 16). We also confirmed that AP scored higher between 0.60 to 0.80 for *Word2vec (Wiki/SNS)+SVM* than the other methods using *SVM*. For samples predicted by the proposed methods, we confirmed that AP scored higher than the aforementioned methods between 0.60 to 1.00 and there was a slight difference between *Word2vec (Wiki/SNS)+BiLSTM+Attention, ELMo+BiLSTM+Attention* and *BERT (Wiki/SNS)+fine-tuning*. That is, AP of samples predicted by *BERT (Wiki/SNS)+fine-tuning* were distributed more often between 0.60 to 1.00 than the other prediction methods using *BiLSTM+Attention* (comparison of (7-9) and (10,11) in both Figs. 15 and 16).

6.3. Comparison of the proposed prediction models

According to the results of recommendation accuracy (see Table 10) and MAP (see Table 14), we ranked the prediction models in the order shown in Table 15. From these results, it can be concluded that :

- Fine-tuning the training data with the BERT model achieved the best results. We assume that using a pre-trained model that can represent the contextual meaning of words works better than a pre-trained model like Word2vec which provides the same vector to a given word.

Table 13
Examples of sentences containing emotive expression, 恥ずかしい (*hazukashii*, "embarrassed").

Data	Text	Emoticon	Emotion type
Training	そう言われると恥ずかしい (*I feel* ***embarrassed****if you say so.*)	(´Д｀ ;)	*Dislike*
Training	見られてたとは恥ずかしい (*You were looking at me, that's embarrassing*	(^_^;)	*Shyness*
that's ***embarrassing****.*)	*that's* ***embarrassing****.*)	(*_*;)	*Shyness*
Test	身近な人に見られるとめっちゃ恥ずかしいです (*I feel very* ***embarrassed****if someone I know sees me)*	(ヽ(；▽；)ノ)	*Sadness*
Test	ずっと見られると恥ずかしい (*If someone looks at me for a while, I feel embarrassed)*	(T_T)	*Sadness*
Test	手相見られてたなんて何だか恥ずかしいです (*It's kind of* **embarrassing to have my palm lines being looked at**	(*ﾉ∀ﾉ))	*Shyness*

Table 14
MAP of each prediction model. "Average" means the average MAP of the entire samples. The highest values are highlighted in bold. (1) *Frequency*, (2) *BoW* + *SVM*, (3) *TFIDF* + *SVM*, (4) *WordCharGram* + *SVM*, (5) *Word2vec (Wiki)* + *SVM*, (6) *Word2vec (SNS)* + *SVM*, (7) *Word2vec (Wiki)* + *BiLSTM* + *Attention*, (8) *Word2vec (SNS)* + *BiLSTM* + *Attention*, (9) *ELMo* + *BiLSTM* + *Attention*, (10) *BERT (Wiki)* + *fine-tuning*, (11) *BERT (SNS)* + *fine-tuning*.

			Emotion type of emotive expression in inputs (with number of samples)										
Method	Polarity Emotion	Average MAP	*Joy* 1930	*Like* 1494	*Relief* 942	*Anger* 420	*Dislike* 2047	*Fear* 812	*Sadness* 947	*Amazement* 588	*Excitement* 720	*Shyness* 324	NONE 9708
(1)	polarity	0.39	0.49	0.44	0.39	0.22	0.38	0.27	0.34	0.22	0.19	0.28	0.39
	emotion	0.22	0.29	0.25	0.23	0.12	0.19	0.16	0.27	0.15	0.14	0.19	0.23
(2)	polarity	0.47	0.50	0.50	0.50	0.42	0.44	0.43	0.45	0.44	0.43	0.45	0.48
	emotion	0.27	0.30	0.29	0.31	0.23	0.22	0.23	0.29	0.27	0.28	0.29	0.27
(3)	polarity	0.47	0.50	0.50	0.50	0.42	0.44	0.43	0.45	0.45	0.43	0.45	0.48
	emotion	0.27	0.30	0.29	0.31	0.23	0.22	0.24	0.29	0.28	0.28	0.29	0.27
(4)	polarity	0.47	0.50	0.49	0.50	0.42	0.44	0.43	0.45	0.44	0.44	0.45	0.48
	emotion	0.27	0.30	0.29	0.31	0.23	0.22	0.24	0.28	0.27	0.28	0.29	0.27
(5)	polarity	0.48	0.51	0.50	0.51	0.43	0.45	0.43	0.45	0.46	0.44	0.47	0.48
	emotion	0.28	0.31	0.30	0.31	0.23	0.22	0.24	0.28	0.28	0.28	0.29	0.28
(6)	polarity	0.49	0.52	0.51	0.52	0.44	0.46	0.44	0.46	0.45	0.45	0.47	0.49
	emotion	0.28	0.31	0.30	0.32	0.24	0.23	0.24	0.30	0.28	0.29	0.30	0.28
(7)	polarity	0.51	0.54	0.54	0.54	0.44	0.49	0.47	0.49	0.47	0.46	0.49	0.51
	emotion	0.30	0.33	0.32	0.33	0.24	0.24	0.25	0.33	0.29	0.29	0.31	0.30
(8)	polarity	0.51	0.55	0.55	0.55	0.44	0.48	0.45	0.49	0.47	**0.47**	0.49	0.51
	emotion	0.30	0.33	0.32	0.34	0.24	0.24	0.25	0.33	0.29	0.30	0.31	0.30
(9)	polarity	0.51	0.55	0.54	0.54	0.44	0.48	0.46	0.48	0.47	0.46	0.48	0.52
	emotion	0.30	0.33	0.32	0.33	0.24	0.24	0.25	0.32	0.29	0.29	0.30	0.30
(10)	polarity	0.53	**0.58**	0.56	0.57	0.45	0.50	**0.48**	**0.52**	0.48	**0.47**	**0.50**	0.53
	emotion	0.31	0.35	0.33	0.35	0.24	0.25	**0.26**	**0.36**	**0.31**	**0.32**	**0.32**	0.31
(11)	polarity	**0.54**	**0.58**	**0.57**	**0.58**	**0.46**	**0.51**	0.47	**0.52**	**0.49**	**0.47**	**0.50**	**0.54**
	emotion	**0.32**	**0.36**	**0.34**	**0.36**	**0.25**	**0.26**	**0.26**	0.35	**0.31**	0.31	**0.32**	**0.32**

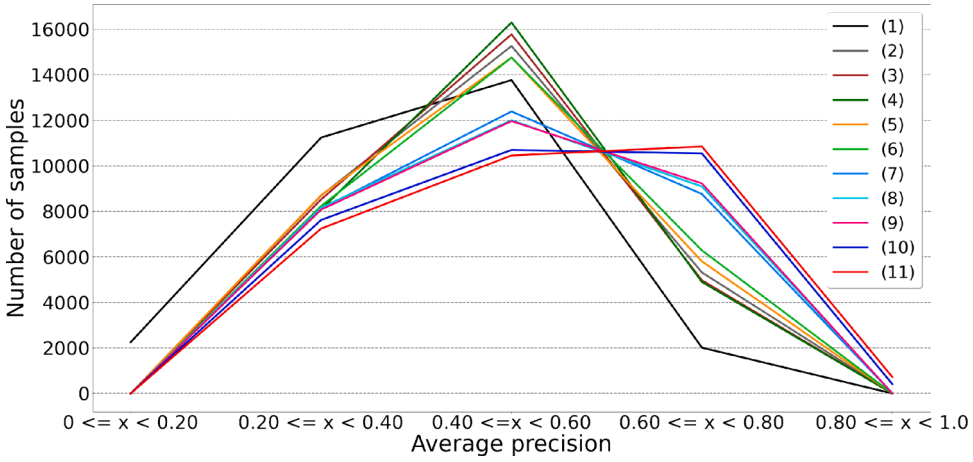


Fig. 15. Distribution of average precision for polarity. Note that samples containing emoticons neither expressing POSITIVE nor NEGATIVE are excluded (see Section 5). (1) *Frequency*, (2) *BoW+SVM*, (3) *TFIDF+SVM*, (4) *WordCharGram+SVM*, (5) *Word2vec (Wiki)+SVM*, (6) *Word2vec (SNS)+SVM*, (7) *Word2vec (Wiki)+BiLSTM+Attention*, (8) *Word2vec (SNS)+BiLSTM+Attention*, (9) *ELMo+BiLSTM+Attention*, (10) *BERT (Wiki)+fine-tuning*, (11) *BERT (SNS)+fine-tuning*.

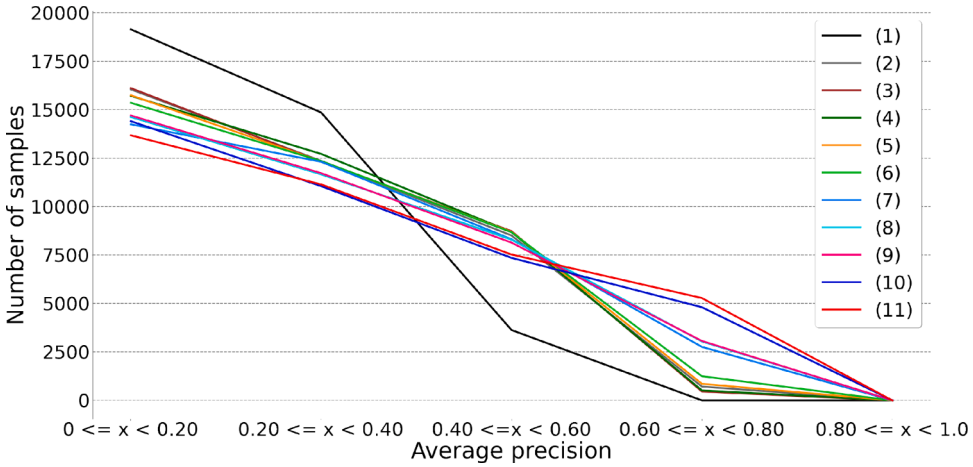


Fig. 16. Distribution of average precision for emotion. (1) *Frequency*, (2) *BoW+SVM*, (3) *TFIDF+SVM*, (4) *WordCharGram+SVM*, (5) *Word2vec (Wiki)+SVM*, (6) *Word2vec (SNS)+SVM*, (7) *Word2vec (Wiki)+BiLSTM+Attention*, (8) *Word2vec (SNS)+BiLSTM+Attention*, (9) *ELMo+BiLSTM+Attention*, (10) *BERT (Wiki)+fine-tuning*, (11) *BERT (SNS)+fine-tuning*.

Table 15
Rank of the prediction models based on the results of recommendation accuracy and MAP.

Method	Recommendation accuracy (%)	MAP (polarity)	MAP (emotion)	Rank
BERT (SNS) + fine-tuning	52.98	0.54	0.32	1
BERT (Wiki) + fine-tuning	50.43	0.53	0.31	2
Word2vec (SNS) + BiLSTM+Attention	48.37	0.51	0.30	3
Word2vec (Wiki) + BiLSTM+Attention	47.63	0.51	0.30	4
ELMo + BiLSTM + Attention	47.42	0.51	0.30	5
Word2vec (SNS) + SVM	45.62	0.49	0.28	6
Word2vec (Wiki) + SVM	44.18	0.49	0.28	7

- Taking the word order and an attention mechanism into account improves the recommendation capability (clearly visible when *Word2vec (SNS/Wiki)+SVM* and *Word2vec (SNS/Wiki)+BiLSTM+Attention* are compared).
- As assumed, models pre-trained on social media corpus achieved better results than the ones originating in Wikipedia articles (compared the results of pre-trained models of *BERT+fine-tuning*, *Word2vec+BiLSTM+Attention*, and *Word2vec+SVM*). We assume

that the way words are used in our training/test data which was collected from blogs and other social media is similar to the social media-based corpus used for the available pre-trained models. Considering the results, we can expect that *ELMo+BiLSTM+Attention* may achieve a better result if we use a pre-trained ELMo model trained on similar corpus instead of one based on news articles.

6.4. Problem of recommending lower rank emoticons

We confirmed that *BERT (SNS)+fine-tuning* model achieved the best results in both recommendation accuracy and MAP, although in 47.0% (17,690 samples) of cases, correct emoticons were not present in the top 25 recommendations made by the system. From our analysis, we were able to determine two reasons. The first one is that the data size for certain emoticons was too small to enter the top 25 ranks. Fig. 17 shows markers indicating the relation between the average ranks of emoticons recommended by each prediction model and the data size of each emoticon in training and test data. X-axis plots emoticon ID ranging from 1 to 167, left Y-axis represents a rank for the markers, and the right Y-axis is data size depicted as lines. As we can see in Fig. 17, the average rank of emoticon declines as the number of training data for the emoticon decreases. Also, 87.0% (15,385 samples) of the 17,690 cases that *BERT (SNS)+fine-tuning* model predicted as correct emoticons from outside of the top 25 had less than 2000 samples in the training data. We assume that the recommendation quality will improve if we can collect more samples for emoticons of smaller data sizes. Although correct emoticons were not recommended at the top due to their infrequent number in the training data, we were able to confirm that our proposed methods recommended emoticons of the same polarity/emotion type with the correct emoticons at the top in the recommendations (Table 16). Table 16 shows the MAP of each prediction model only focusing on emoticons that have less than 200 samples in the test data. As shown in the Table 16, we discovered that our proposed models were better than the conventional methods by 0.01 (difference between *Word2vec (Wiki)+SVM* and *WordCharGram+SVM*) to 0.12 (difference between *BERT (SNS)+fine-tuning* and *Frequency*). These results indicate that the proposed methods can recommend emoticons expressing similar polarity or emotion type of the correct emoticon to input sentences.

The second reason was that the tendency of emoticons used in some sentences was different from similar sentences in the training data, thus, the prediction model could not recommend the correct emoticon within the top 25. This reason could be applied for any kind of sentences (sentences containing or not containing emotive phrases). We extracted 12,067 samples that *BERT (SNS)+fine-tuning* model failed to (a) recommend correct emoticons within top 25 and (b) recommend emoticons of the same polarity/emotion type at the top of the list ($MAP < 0.50$). We analyzed what kind of emoticons are inserted into similar sentences found in the training data. We calculated the similarity of sentences using the Flatten layer (see Fig. 11) extracted from *BERT (SNS)+fine-tuning*model with cosine similarity. Cosine similarity was calculated by Eq. (10) where $v1$ is the vector of the Flatten layer for the first sentence, $v2$ is the vector of the Flatten layer for the second sentence and N is the vector size. For each of the 12,067 samples, we collected sentences in which cosine similarity is higher than 0.90 and calculated the proportion of the emoticons of the polarity/emotion type that is the same as the emoticon in each sample. For example, there is a sentence with an emoticon "(^_^)" that belongs to the Positiveand *Joy* categories and we discovered that there are 100 similar sentences. If there are 37 emoticons that express *Joy* and 67 emoticons that express Positive among 100 sentences, then the proportion will be 37% (emotion type) and 67% (polarity). Therefore, a higher proportion means

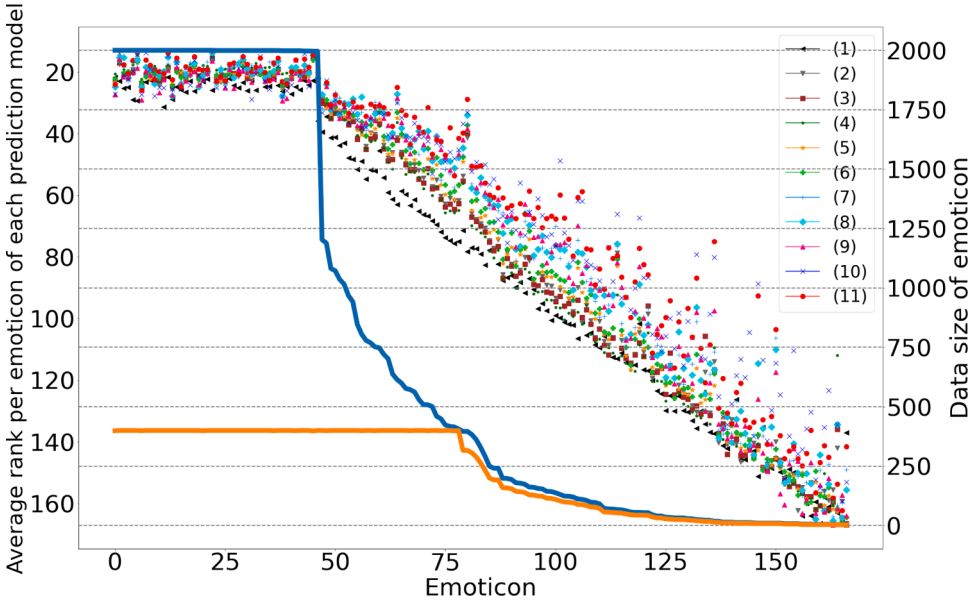


Fig. 17. Average rank per emoticon of each prediction model (represented as markers) and data size of each emoticon in the training/test data (shown as lines). (1) *Frequency*, (2) *BoW+SVM*, (3) *TFIDF+SVM*, (4) *WordCharGram+SVM*, (5) *Word2vec (Wiki)+SVM*, (6) *Word2vec (SNS)+SVM*, (7) *Word2vec (Wiki)+BiLSTM+Attention*, (8) *Word2vec (SNS)+BiLSTM+Attention*, (9) *ELMo+BiLSTM+Attention*, (10) *BERT (Wiki)+fine-tuning*, (11) *BERT (SNS)+fine-tuning*.

Table 16
MAP of each prediction model for emoticons less than 200 samples in the test data. The highest values are highlighted in bold. (1) *Frequency*, (2) *BoW+SVM*, (3) *TFIDF+SVM*, (4) *WordCharGram+SVM*, (5) *Word2vec (Wiki)+SVM*, (6) *Word2vec (SNS)+SVM*, (7) *Word2vec (Wiki)+BiLSTM+Attention*, (8) *Word2vec (SNS)+BiLSTM+Attention*, (9) *ELMo+BiLSTM+Attention*, (10) *BERT (Wiki)+fine-tuning*, (11) *BERT (SNS)+fine-tuning*.

	Method										
Type	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Polarity	0.40	0.46	0.46	0.46	0.47	0.48	0.49	0.50	0.50	0.52	**0.53**
Emotion	0.22	0.25	0.25	0.25	0.25	0.26	0.27	0.27	0.27	**0.29**	**0.29**

emoticons of the same polarity/emotion type are also used in similar sentences in the training data. The results are shown in Table 17. As can be seen in Table 17, an average of 12.6% and 23.8% of emoticons in similar sentences had identical emotion and polarity, respectively, for 12,067 samples that did not recommend correct emoticons from the top 25. On the other hand, an average of 37.4% and 56.2% of emoticons in similar sentences had the same emotion type and polarity, respectively, for 19,932 samples that recommended correct emotions within the top 25. This indicates that similar sentences in the training data for 12,067 samples used emoticons differently. For example, the sentence ``わからなすぎてやる気が出ない *(I have no motivation because I can't get it at all)*" which does not contain any clear emotional statements appeared with ``(^ ∇ ^)" emoticon, expressing *Joy* in the test data, however, the similar sentences in the training data tend to appear with emoticons of *Sadness* or *Dislike* (see Table 18). We believe that there are patterns suggesting how emoticons are used in certain contexts, however, the way emoticons are used with sentences may differ depending on a user, current state, and so on. Therefore, in order to improve our method, combining other aspects such as users' emoticon preferences, or users' preference towards a topic may be useful.

$$Sim(v1, v2) = \frac{\Sigma_{i=1}^{N} v1_i v2_i}{\sqrt{\Sigma_{i=1}^{N} v1_i^2}\sqrt{\Sigma_{i=1}^{N} v2_i^2}} \tag{10}$$

7. Conclusion

In this paper, we presented an emoticon recommendation system that reorders an emoticon list regarding the entire context of input to support users' emoticon selection. This was to tackle the problem of previous methods relying on lexicons. Our approach was to create a prediction model utilizing machine learning on sentence-emoticon pairs with pre-trained models trained on large data in Japanese. We implemented four prediction models and evaluated which one can recommend users' desired emoticons with the highest accuracy. The four prediction models were: Word2vec model with SVM classifier, Word2vec model with attention-based BiLSTM, ELMo model with attention-based BiLSTM, and fine-tuning with BERT model. In addition, we evaluated the influence of differences in data domains used for pre-training by comparing pre-trained Word2vec and BERT models trained on social media corpus and Wikipedia articles. We used 124,359 data samples for training and 37,622 data samples for testing. The training and test data contained sentence-emoticon pairs using 167 unique emoticons.

The results of our experiments showed that the four prediction models achieved results ranging from 44.18% to 52.98% in recommending correct emoticon within the top 25 and the best method was to use fine-tuning with BERT model. These results were better than those of conventional methods, recommending emoticons based on frequency or using Bag-of-Words or word and character n-grams as features with SVM which ranged from 29.53% to 43.58%. We also confirmed our intuition that using a pre-trained model that trained on social media corpus achieved slightly better results (by 0.74 pt to 2.55 pt) than the model trained on Wikipedia articles.

For data with lower recommendation results, we learned that more training data is required for emoticons with a small number of samples to achieve a better result. Moreover, some input sentences were not able to output correct emoticons because polarity and emotion type of emoticons used in similar sentences inside the training data were different. This means that there are patterns of how emoticons are used with messages but human-selected emoticons may differ based on the user's preferences (e.g. user may tend to choose crying emoticon in happy context, "tears of joy") and topics (e.g. John likes Pokémons so he inserts happy emoticon inside sentences related to the game.). Therefore, we believe that adding other information such as users' preferences could be beneficial for further improvement of our system and be more personalized to each user in the real-world applications of input methods.

Table 17
Average proportion of emoticons of the same polarity/emotion type used in similar texts in training data.

Comparison	Average	
	Polarity	Emotion
12,067 samples ($rank > 25$& $MAP < 50$)	23.8%	12.6%
19,932 samples ($rank <= 25$)	56.2%	37.4%

Table 18
Examples of texts from training data where emoticon usage is different from the sample in the test data (the first sentence in bold is the source for finding similar sentences).

Comparison	Data	Text	Emoticon	Emotion type
examples from 12,067 samples	**Test**	わからなすぎてやる気が出ない (***I have no motivation because I can't get it all)***	(^▽^)	***Joy***
	Training	なんもやる気が出ない (*I have no motivation*)	(ー _ ー ;)	*Dislike*
	Training	あたしもやる気出ない (*I don't have motivation either*)	(¯¯;)	*Dislike*
	Training	やる気が起きないご (*I don't feel like doing anything*)	(T T)	*Sadness*
	Training	掃除したいけど寒くてやる気起きません (*I want to clean the room but I have no motivation because it's cold*)	(; ;)	*Sadness}*
	Training	今日は何だか何もやる気起きないんだ(*I have no motivation today.*)	(´・_・)	*Sadness*

CRediT authorship contribution statement

Yuki Urabe: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Writing - review & editing, Visualization. **Rafal Rzepka:** Methodology, Validation, Resources, Writing - review & editing, Supervision. **Kenji Araki:** Validation, Writing - review & editing, Supervision.

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